

Reverse predictivity for bidirectional comparison of neural networks and biological brains

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A major goal in systems neuroscience is to build computational models that capture the primate brain's internal representations. Standard evaluations of artificial neural networks (ANNs) emphasize forward predictivity—how well model features predict neural responses—without testing whether model representations are themselves predictable from neural activity. Here we develop a diagnostic metric, reverse predictivity, that quantifies how well macaque inferior temporal cortex responses predict ANN unit activations. Using this comparative framework, we reveal a striking asymmetry: models with high forward predictivity (~50% variance explained) often contain units unpredictable from neural activity, reflecting biologically inaccessible dimensions. In contrast, monkey-to-monkey mappings are symmetric, providing an empirical reference point and indicating that the asymmetry reflects genuine representational mismatch. Reverse predictivity enables the identification of ‘common’ ANN units that are shared with the inferior temporal cortex, are behaviourally relevant and generalize across species, and ‘unique’ units lacking such alignment. Influenced by feature dimensionality, training objectives and adversarial robustness, reverse predictivity serves as a conservative diagnostic and comparative tool for guiding next-generation ANNs towards both high task performance and greater biological plausibility.

Over the past decade, deep artificial neural networks (ANNs) trained on visual tasks have transformed our understanding of primate sensory systems^{1,2}. These computational models, optimized primarily for image recognition tasks, have emerged as powerful hypothesis-generation frameworks for neural processing in the primate ventral visual stream, particularly within the inferior temporal (IT) cortex^{1,3}. Seminal studies have demonstrated striking similarities between ANN internal representations and neuronal activity in macaque V1⁴, V2⁵, V4^{1,6,7} and IT^{1,6,8} cortices, as well as human visual^{9–12}, auditory^{13,14} and language^{15,16} areas, underscoring their utility in neuroscience. Indeed, state-of-the-art ANNs consistently account for roughly half of the explainable variance in IT neural responses⁵,

reflecting their ability to capture essential aspects of the brain's representational structure.

Despite these promising results, substantial gaps remain in fully understanding the extent and nature of model–brain alignment. The conventional metric^{1,5} for evaluating model–brain correspondence—referred to here as forward predictivity—focuses solely on how well ANN features predict neural responses through a permissible linear transformation. Forward predictivity, therefore, is inherently a one-way comparison: it tests how well model units predict neurons, but not the converse. If an ANN were a perfect model of the IT cortex, the correspondence should be bidirectional—one ought to be able to map from model units to neurons and from neurons to model

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units with comparable fidelity. This bidirectionality provides a useful diagnostic expectation, yet this assumption has rarely been tested explicitly within ANNs or primates. Previous efforts^{17,18} have typically optimized models to maximize forward predictivity, implicitly treating the brain-to-model mapping as given. There has been growing recognition that two representational spaces might align well in one direction yet differ in dimensionality or information content^{19–21}. In other words, ANN models and brains might not share all features, leading to an asymmetry in their relationship that forward metrics alone cannot reveal²².

Models may contain many units that bear little correspondence to neural responses yet still achieve high forward predictivity because a smaller subset aligns well with the IT cortex. Thus, forward predictivity alone cannot reveal how much of a model's representation truly overlaps with the brain. To address this, we introduce reverse predictivity, which asks the complementary question: to what extent can IT neurons predict the activations of individual ANN units? This metric provides a diagnostic and comparative assessment of how much of the model's feature space lies within the linear span of the neural population. Units that fall within this shared subspace should be predictable from the IT cortex, whereas units outside it, not used by the brain, will show low predictability, revealing representational asymmetry. While neural and model representations could in principle be related through nonlinear transformations not captured by linear mappings, we emphasize that low reverse predictivity does not imply the biological absence of a representational dimension. Rather, it indicates that such a dimension is not linearly recoverable from the sampled IT population under current measurement constraints. Our use of linear bidirectionality is guided by an empirical benchmark: when the identical linear framework is applied between two monkeys' IT populations, forward and reverse predictivity are closely matched. If an ANN were a representational analogue of the IT cortex at the resolution accessible to our recordings, it should show comparable symmetry. Departures from this monkey–monkey symmetry therefore suggest representational mismatch at least at the linear level, even while nonlinear correspondences remain theoretically possible.

Using large-scale neural recordings from macaque IT cortex⁸, we systematically probe representational asymmetries across a diverse suite of ANN architectures, including convolutional networks (for example, AlexNet²³), deeper feedforward convolutional neural networks (CNNs; for example, ResNet-50²⁴, Inception-v3²⁵), vision transformers (for example, ViT²⁶) and modern convolutional designs (for example, ConvNeXt²⁷) with a two-way benchmark that complements forward predictivity (ANN to brain) with reverse predictivity (brain to ANN). Within this framework, we distinguish common units whose activations are reliably predictable from the IT cortex and consistently support neuron prediction, from unique units, whose activations show little alignment with neural activity. To biologically calibrate the benchmark, we compare monkey-to-ANN mappings with monkey-to-monkey mappings, in which one animal's IT population predicts the other's. The latter are markedly more symmetric, confirming that the observed asymmetries are characteristic of model-to-brain, not brain-to-brain comparisons. Reverse predictivity exposes substantial mismatches that forward metrics alone do not reveal, and it varies systematically with factors such as effective dimensionality²⁸, adversarial robustness²⁹ and ImageNet accuracy³⁰, often leading to divergent inferences from forward versus reverse evaluations. Collectively, these results motivate a bidirectional diagnostic standard for model–brain assessment: reverse predictivity serves both as a diagnostic of a potential hidden misalignment and as a design target, guiding the reduction of extraneous unique dimensions and promoting ANN representations that are more biologically plausible and behaviourally predictive.

Results

As outlined above, complementary to the standard practice of predicting neurons from ANNs^{1,5,8}, in this study, we probe how symmetric this

relation is across models and monkeys. Specifically, as detailed in the sections below, we highlight the limitations of inferring ANN–brain alignment from forward predictivity and augment this metric with a diagnostic analysis that asks how well monkey neurons recorded across the IT cortex can predict ANN–IT cortex units.

Forward predictivity masks differences in model–brain alignment

The dominant approach for comparing ANNs with biological vision systems is forward predictivity—the degree to which model features can be used to predict neural responses^{1,8}. This technique (schematized in Fig. 1a) involves learning a linear mapping from model features to neural activity and has become the standard benchmark for evaluating model–brain alignment⁵. Yet as more high-performing models are developed, often optimized for classification benchmarks such as ImageNet³⁰, their forward predictivity scores have begun to plateau⁵ (Extended Data Fig. 1). Across architectures—from convolutional networks to transformers—percentage explained variance (%EV) in the macaque IT cortex typically hovers around 50% (Extended Data Fig. 1). To illustrate an important caveat that is present within these seemingly equal high forward predictivity, we show the comparison of two representative ANNs: DenseNet-161³¹ and VGG16³². For both models, we extracted features from an IT-aligned layer (based on Brain-Score⁵; see ‘Layer selection’ in Methods), trained a linear regression model to predict the neural responses (monkey 1, 169 sites; monkey 2, 107 sites) from these features, and computed a cross-validated, noise-corrected %EV (‘Forward predictivity analysis’ in Methods) for each recorded macaque IT neuron. Both models achieved nearly identical performance (DenseNet-161, median = 54% ± 10.74 (median absolute deviation (MAD))); VGG16, median = 53% ± 10.22 (MAD)), with overlapping distributions of %EV (Fig. 1b,c; $t(212) = -0.815$, $P = 0.416$). At first glance, these results suggest that the models are similarly brain-like. However, dissecting how they achieve their performance revealed stark differences. As shown schematically in Fig. 1a, DenseNet-161 contains fewer units that do not contribute to the regression compared with VGG16 (which contained a larger set of non-contributing units). This was confirmed by comparing the distributions of normalized regression weights (Fig. 1d,e). We observed that the median normalized regression weight for VGG16 (median = 0.09 ± 0.05 (MAD)) was lower compared with DenseNet-161 (median = 0.24 ± 0.08 (MAD)). A Wilcoxon rank-sum one-tailed test confirmed the statistical difference between both distributions ($Z = -252.105$, $P < 0.001$). Given these observations, we hypothesized that units with consistently strong contributions across neurons reflect common representational components shared with the brain, whereas units with near-zero weights encode more unique dimensions (idiosyncratic to the ANNs). Thus, while forward predictivity captures how well a model predicts neural activity, it does not reveal which units drive that performance. Two models can have identical forward predictivity scores yet rely on fundamentally different internal strategies—differences that are biologically distinct and might be functionally meaningful.

Reverse predictivity reveals asymmetries between models and brains

To address the limitations of forward predictivity, we introduced a complementary diagnostic metric: reverse predictivity. Rather than asking whether model features can predict neural activity, we asked whether neural responses could predict the activations of individual model units (Fig. 2a). This reverses the direction of inference and probes whether a model's internal features are linearly predictable from brain activity. We implemented reverse predictivity by training a linear regression model for each unit in a model's IT-aligned layer, using IT responses as input (see ‘Reverse predictivity analysis’ in Methods). The model's reverse predictivity score was defined as

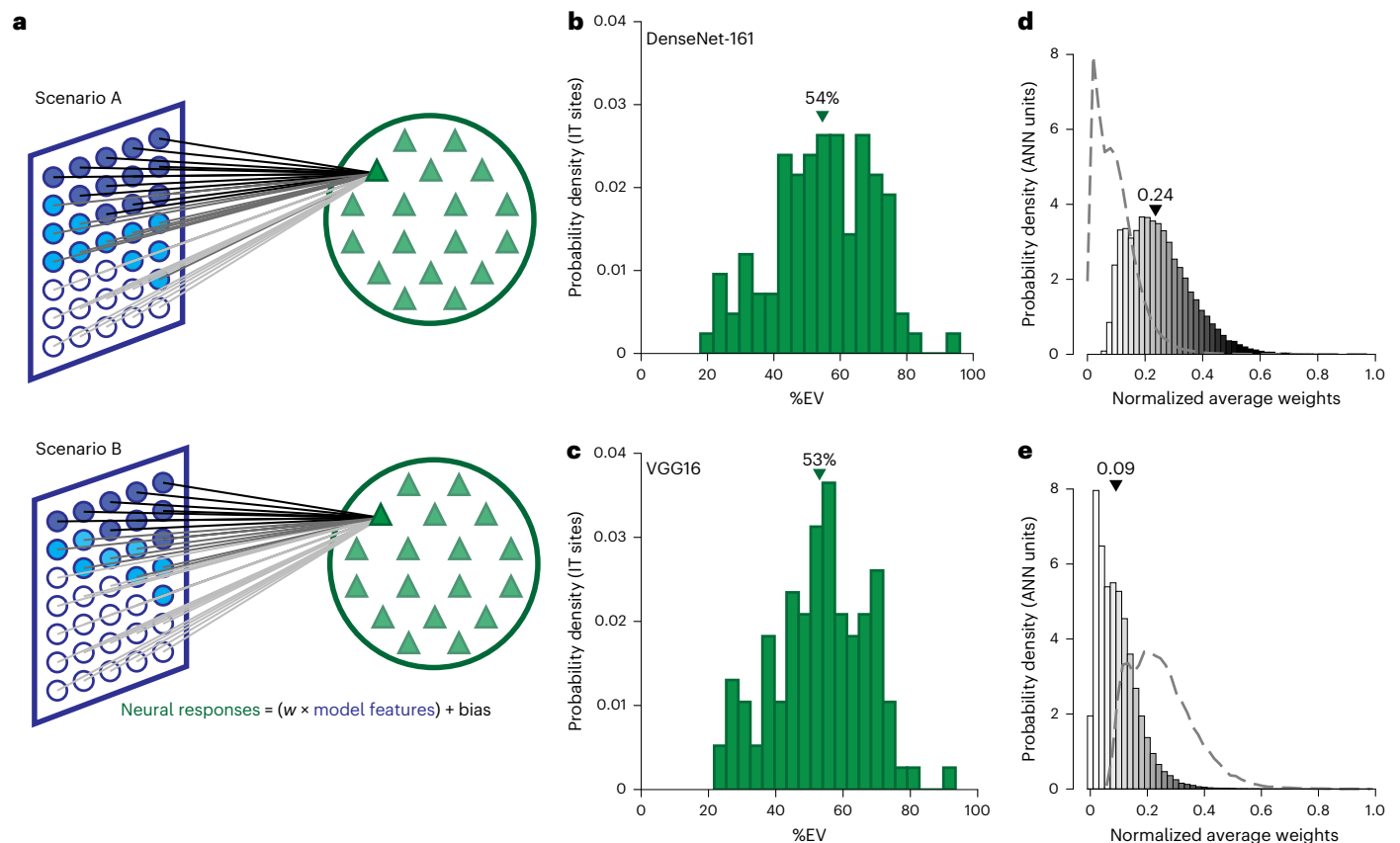


Fig. 1 | Similar predictive performance, different underlying strategies across models. **a**, Schematic illustration of two scenarios for mapping ANN features (left) to neural responses (right), highlighting differences in unit contribution. Line darkness indicates the contribution strength of the learned linear mapping (reflected in the weights w of a linear regression equation shown below) from a model unit to a neuron. Scenario A (for example, DenseNet-161) shows that many units are contributing, while scenario B (for example, VGG16) exhibits more

non-contributing units. **b,c**, Distributions (shown as probability density plots) of %EV across macaque IT neurons when predicted by DenseNet-161 (**b**) or VGG16 (**c**) features. **d,e**, Distributions (shown as probability density plots) of normalized averaged regression weights assigned to each model unit for DenseNet-161 (**d**) and VGG16 (**e**). The dashed lines indicate the weight distributions of VGG16 (**d**) and DenseNet-161 (**e**).

the median %EV across its units. If the differences in forward mapping weights discussed earlier genuinely reflect representational misalignment with the brain, we would expect ANNs that contain more units with higher forward weights to also be more predictable from IT responses (that is, a higher reverse predictivity), and vice versa. To test this, we compared the average reverse predictivity scores to the average forward regression weights across ANNs (Fig. 2b). Overall, we observed a significant positive correlation between the two measures (Spearman $r(62) = 0.4$, $P = 0.001$; Fig. 2b). This relation was much stronger, when we looked at ANNs clustered according to their architectures. Specifically, we observe a strong correlation within convolutional models (Spearman $r(46) = 0.51$, $P < 0.001$), suggesting that within well-matched architectures, reverse predictivity provides a reliable means of identifying which model features are most functionally aligned with neural representations under a linear mapping assumption. Consequently, reverse predictivity offers a more direct, unit-centric assessment of model–brain alignment, enabling us to classify individual units as ‘common’ (high reverse %EV) or ‘unique’ (low or near-zero reverse %EV), based on whether they contribute to forward neural predictivity and are themselves predictable from IT responses. To further clarify the functional contributions of common and unique ANN units, we performed a feature-ablation analysis (Extended Data Fig. 2). For each model, we first identified common and unique units using an independent set of images and neural responses to maintain proper cross-validation. We then selectively removed either the common or the unique units from the ANN–IT layer

and recomputed forward and reverse predictivity. Across models, predictivity after removing unique units was consistently higher than predictivity after removing common units for both forward and reverse mapping (paired tests: $Z = 603$, $P < 0.001$; $Z = 0$, $P < 0.001$). These results further emphasize the functional relevance of the common subspace: ablating common units reliably reduces both forward and reverse predictivity, whereas ablating unique units preserves forward predictivity while increasing reverse predictivity, indicating that common units disproportionately support neural prediction rather than merely scoring highly under reverse predictivity. Thus, common units carry the dimensions that are jointly aligned with the IT cortex, while unique units disproportionately contribute to the observed asymmetry.

Returning to DenseNet-161 and VGG16, we observed a clear divergence in reverse predictivity. Despite similar forward scores, DenseNet-161 (Fig. 2c) showed a broad distribution of reverse %EV, while VGG16’s distribution (Fig. 2d) was heavily skewed towards zero. This difference was highly significant (Wilcoxon rank-sum one-tailed test: $Z = -252.105$, $P < 0.001$), with average reverse predictivity score substantially higher for DenseNet-161 (median = 16.87, MAD = 5.96) than for VGG16 (median = 5.66, MAD = 2.57). These findings are consistent with the forward weight analyses and confirm that similar forward predictivity scores can mask deep representational disparities between models and brains.

We next computed forward and reverse predictivity scores for a diverse set of architectures—covering CNNs, transformers,

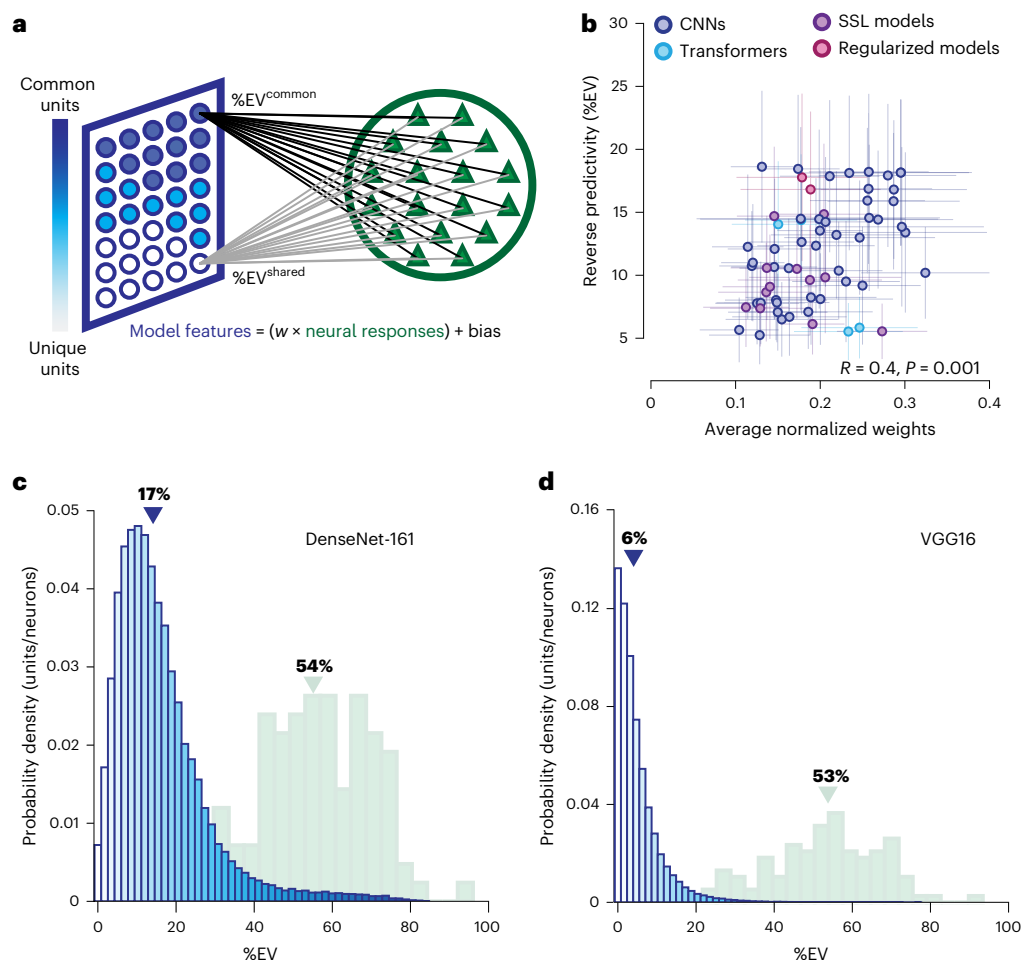


Fig. 2 | Reverse predictivity reveals differences in model-to-brain mapping strategies. **a**, Reverse predictivity quantifies how well individual model units can be predicted from neural responses. Units with high reverse predictivity are considered ‘common’, while low reverse predictivity indicates ‘unique’ units. **b**, Reverse predictivity versus average normalized weights across ANN models

(CNNs, transformers, and SSL and regularized models). Error bars indicate the median absolute deviation across the number of units in each model (see the numbers in Extended Data Fig. 10). Statistics were computed using Spearman correlation across 64 models. **c,d**, Comparison of forward (green) and reverse (blue) predictivity distributions for DenseNet-161 (**c**) and VGG16 (**d**).

self-supervised models and adversarially robust variants (see ‘Model selection’ in Methods and Extended Data Fig. 10 for the list of models), using two different image sets and neural recordings (two monkeys; monkey 1, 130 sites; monkey 2, 74 sites; ‘Visual stimuli’ in Methods).

On both datasets, we observed two striking results. First, reverse predictivity scores are significantly lower than the corresponding forward predictivity scores (Wilcoxon signed-rank one-tailed test: $P < 0.001$). Second, we observed that forward and reverse predictivity were not positively correlated across models (Fig. 3a, Pearson $r(76) = -0.31, P = 0.005$; Fig. 3b, Pearson $r(76) = 0.16, P = 0.18$). We speculate that strategies that over the course of time have yielded higher ImageNet accuracies, thereby improving forward predictivity (to a certain limit), might have introduced large mismatches between the representational structures of these ANNs with the brain—not yet accessible with forward predictivity-based metrics. However, such asymmetries might be a feature of complex systems. Therefore, to first rule out that this asymmetry was not common to all area-to-area comparisons across brains or machines, we applied the same analysis to monkey–monkey mappings. Interestingly, when using one monkey’s IT activity to predict the other’s, forward and reverse predictivity were closely matched (green dots in Fig. 3), with scores along the unity line (Fig. 3a; monkey 1 predicting monkey 2 yielded: forward predictivity, median = 35.5 ± 7.97 (MAD); reverse predictivity, median = 43.2 ± 10.5 ; and vice versa). We note that these monkey–monkey comparisons

serve primarily as empirical reference points within the scope of these analyses. This symmetry supports the interpretation that the observed model–brain asymmetry reflects a genuine representational mismatch. To examine whether the mismatch in dimensionality between ANN layers and the number of sampled IT neurons could account for this effect, we repeated the analysis after down-sampling each model’s IT-layer features to match the number of recorded neurons (288 components), using principal components identified from a held-out image set. This dimensionality-matched comparison produced the same qualitative pattern of model–brain asymmetry alongside preserved monkey–monkey symmetry, indicating that the observed asymmetry cannot be attributed to trivial differences in feature count (Extended Data Fig. 3). In addition, model-to-model comparisons revealed substantially greater symmetry (across different initializations or architectures) than model-to-monkey comparisons (Extended Data Fig. 4), further indicating that the observed asymmetry is specific to ANN–brain representational differences rather than sampling or scaling limitations. Throughout, linear bidirectionality is intended as a conservative diagnostic criterion for identifying representational mismatch between ANNs and brains. Finally, similar results were obtained when using a one-to-one mapping procedure³³ in place of the standard many-to-one regression (Methods and Extended Data Fig. 5).

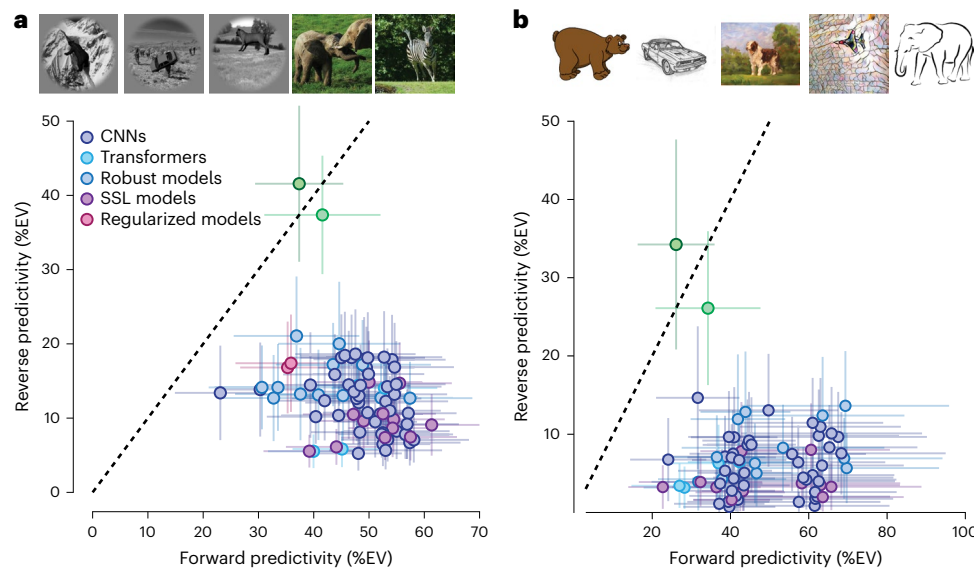


Fig. 3 | Reverse and forward predictivity do not align across models. a, Top: example of images from our datasets. Bottom: forward versus reverse predictivity metrics. Each dot represents a model–monkey pair, spanning a range of architectures and training regimes (CNNs, transformers, and SSL, robust and regularized models). The green dots denote monkey-to-monkey predictivity, and blue to red coloured dots correspond to ANN-to-monkey

(forward) versus monkey-to-ANN (reverse) predictivity. **b,** The same as in **a** but on a different set of images. The error bars represent the median absolute deviation across model units on the y axis (see the numbers in Extended Data Fig. 10) and monkey neurons on the x axis ($N_1 = 169$ and $N_2 = 107$). Images adapted from: **a**, greyscale images, ref. 35 under a Creative Commons licence CC BY 4.0; coloured photographs, ref. 44, Springer; **b**, ref. 67.

Model architecture and training influence representational alignment with the brain

Next, we asked whether certain choices in model development determine a model's reverse predictivity. Therefore, we examined how certain architectural choices, the dimensionality of the ANN-IT layers and the training objectives affect the macaque IT's predictivity of the ANN's units under this diagnostic framework. First, we found that while the total parameter count of the ANNs (see 'Layer selection' in Methods) had no significant effect (Fig. 4b; Spearman $r(76) = -0.15$, $P = 0.237$), models with more units in their IT-aligned layer tended to have lower reverse predictivity (Fig. 4a; Spearman $r(76) = -0.28$, $P = 0.012$). This suggests that feature width—not overall model size—could drive the emergence of brain-misaligned units. Second, we examined each model's effective dimensionality²⁸—quantified as the participation ratio of the principal component analysis (PCA) eigenvalue spectrum, which reflects how many dimensions are effectively used to represent visual information (see 'Effective dimensionality' in Methods). Consistent with previous results²⁸, we indeed observed that forward predictivity increased with effective dimensionality (Fig. 4c; Spearman $r(76) = 0.22$, $P = 0.05$). In contrast, reverse predictivity decreased with effective dimensionality (Fig. 4d; Spearman $r(76) = -0.53$, $P < 0.001$), indicating that increasingly high-dimensional representations contain features not linearly recoverable from IT. Third, we asked whether learning strategies that are known to reduce the dimensionality of ANN-IT features, yielding a better representational match with macaque IT neurons³⁴, could improve reverse predictivity. Therefore, we tested the effects of adversarial training to generate more robust models on reverse predictivity. Indeed, adversarially robust models consistently outperformed their non-robust counterparts in reverse predictivity (Fig. 4e; $t(11) = 5.762$, $P < 0.001$). Thus, robust training appears to regularize models towards more biologically compatible features. Finally, we further explored the role of learning strategies during model training. Comparing supervised learning and self-supervised learning (SSL) models with identical architectures, we observed a non-significant higher average reverse predictivity in supervised models (Fig. 4f; $t(11) = 2.07$, $P = 0.06$). While not statistically significant, the

effect suggests that SSL may induce less brain-aligned representations. In addition, while we find a positive correlation between forward predictivity and ImageNet³⁰ top-1 accuracy (Spearman $r(76) = 0.46$, $P < 0.001$; Extended Data Fig. 4a), we observed a significant negative correlation between reverse predictivity and ImageNet top-1 accuracy (Spearman $r(76) = -0.3$, $P = 0.012$; Extended Data Fig. 4b). Together, these results show that reverse predictivity is sensitive to certain ANN design choices, and these influences are not necessarily aligned with improvements in forward predictivity or task accuracy.

Unique units reflect genuine representational mismatches not sampling limitations

Could unique units simply reflect noisy data or limitations in neural sampling during recordings? To rule this out, we first tested whether ANN unit predictability was consistent across animals. We computed reverse predictivity for each ResNet-50 unit using two monkeys separately. The results were highly correlated (Spearman $r(100,350) = 0.87$, $P < 0.001$), suggesting a strong agreement in unit ranking (Fig. 5a; see also Extended Data Fig. 5b for consistent correlations across all tested models). Next, we tested whether low predictivity arose from under-sampling. Focusing on the 500 least predictable ResNet-50 units, we subsampled neurons from 20 to 300 and assessed reverse predictivity (Extended Data Fig. 5c; see also 'Sampling analysis of reverse predictivity' in Methods). Performance plateaued quickly (with maximal reverse predictivity scores of $2.87\%EV \pm 0.07$ (MAD) for monkey 1 and $2.85\%EV \pm 0.06$ (MAD) for monkey 2), suggesting that even with more neural samples, these units remain linearly unrecoverable from the population of macaque IT cortex. The responses of the unique units, however, were not inherently unpredictable. They were highly predictable from other model units—especially within the same model instance (Fig. 5c, dark blue dots), but also from other instances (pre-training weight initializations) of the same model (Fig. 5c, lighter blue dots). We observed that predictability of the unique units was lower for different random initializations of the same ANN architectures compared with the same instance (same instance %EV > different instances %EV Wilcoxon signed-rank one-tailed test: $Z = 36.0$, $P < 0.001$),

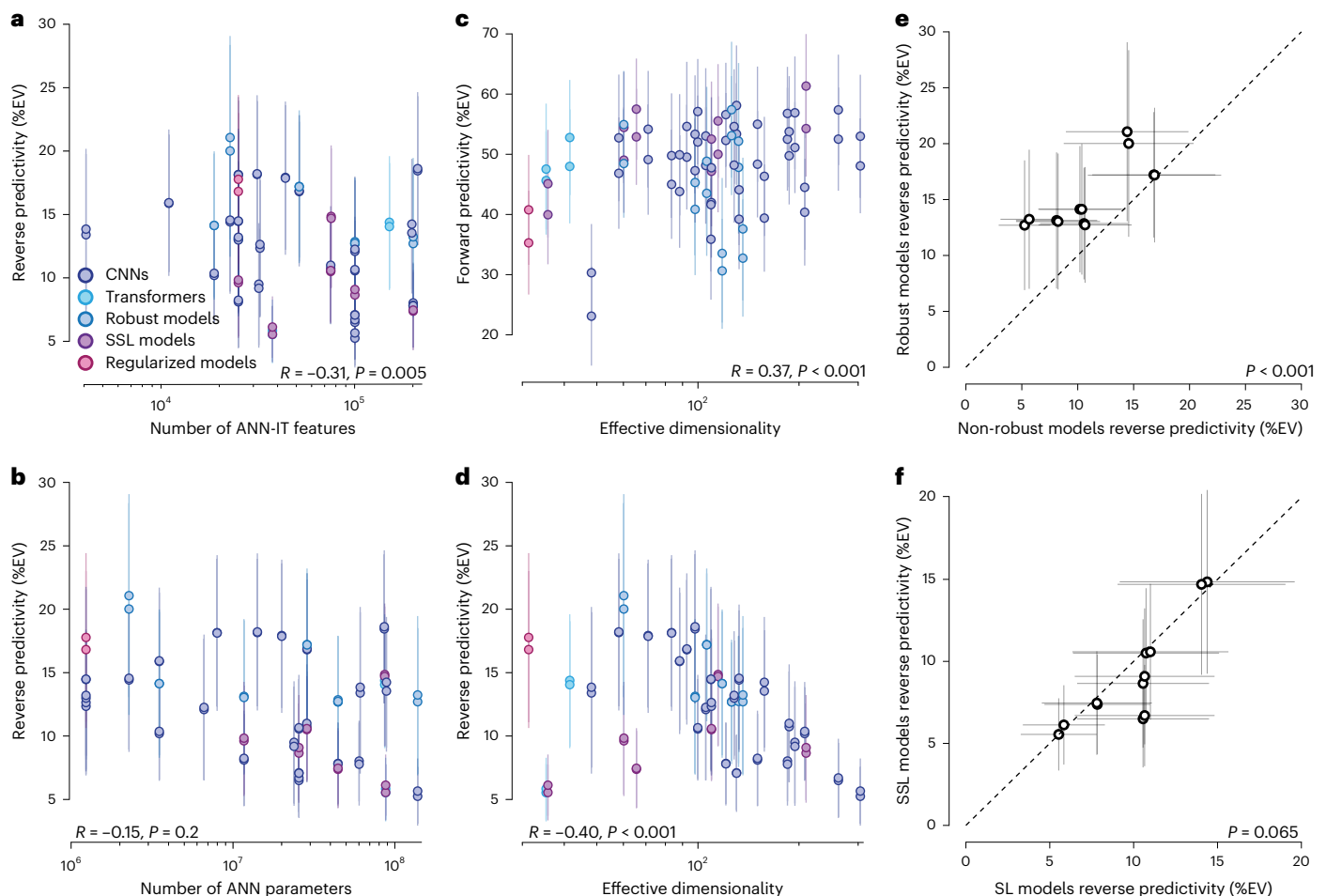


Fig. 4 | Factors influencing reverse predictivity. **a, b**, Reverse predictivity across models as a function of the number of features in the IT layer used for mapping to neural data (**a**) and total number of model parameters (**b**). **c, d**, Forward (**c**) and reverse (**d**) predictivity against effective dimensionality of model representations ($P = 0.00084$ (**c**); $P = 0.00030$ (**d**)). **e**, Reverse predictivity for robust versus non-robust models, matched in architecture. **f**, Same comparison for SSL versus supervised learning (SL) model pairs. In **c**, error

bars represent the median absolute deviation across monkey neurons on the x axis ($N_1 = 169$ and $N_2 = 107$). In **a**, **b** and **d–f**, the error bars represent the median absolute deviation across units per model (see the numbers in Extended Data Fig. 10). For **a–d**, statistics were computed using Spearman correlation across 78 model–monkey pairs; for **e** and **f**, two-sided statistical test across 12 model–monkey pairs, no adjustments for multiple comparisons (paired Wilcoxon test, $P = 0.00049$ (**e**); paired t -test, $P = 0.065$ (**f**)).

indicating that these unique units encode structured but model-specific representations. This result was consistent across models with units within models being better predictive of unique units than macaque IT neurons (Extended Data Fig. 4d).

Common units across models are better predictors of downstream behavioural variance across species

To investigate the functional significance of the common and unique units, we next computed the ability of these units to predict object discrimination behaviour across various models and primates. We selected the top (common) and bottom (unique) 20% of ANN-IT units based on reverse predictivity and applied a subsampling procedure to ensure that common and unique units reach comparable accuracies ('Consistency of common and unique units with behavioural responses' in Methods and Extended Data Fig. 6). Then we trained cross-validated linear classifiers on responses from each unit type to decode object identity, generating image-by-image behavioural accuracies (Fig. 6a and 'Behavioural metrics' in Methods). These predictions were compared with those from other models and with empirical behavioural responses from 3 monkeys and a pool ($n = 88$) of human participants (Fig. 6b and 'Consistency of common and unique units with behavioural responses' in Methods). Because common units are selected

based on their alignment with IT cortex responses—a region that both supports core object recognition and robustly predicts behavioural performance³⁵—they are expected to be more behaviourally relevant, reflecting task-relevant visual representations.

In all comparisons, common units outperformed unique ones. Their behavioural predictions were more consistent with other models (Fig. 6c; Wilcoxon signed-rank test: $Z = 3,081.0, P < 0.001$) with common units achieving higher consistency (median = 0.43 ± 0.009) compared with unique units (median = 0.35 ± 0.004). Interestingly, common units are also more aligned with primates' (both monkeys and humans) behaviour (Fig. 6d; $t(7) = 5.71, P < 0.001$). This trend held across architectures (Extended Data Fig. 6) and when common and unique units were selected based on one-to-one mapping (Extended Data Fig. 7). Furthermore, common ANN units, identified using neural activity from two monkeys separately, better predicted behaviour in other monkeys and humans than unique units (Fig. 6e; $t(7) = 6.24, P < 0.001$). In addition, selectively removing common units in the ANN-IT layer impacted the model's consistency with primates' behaviour (Extended Data Fig. 8) more compared with removing unique units. Thus, reverse predictivity serves as a diagnostic means of isolating units that not only align with IT but also disproportionately support forward behavioural predictivity and cross-species generalization.

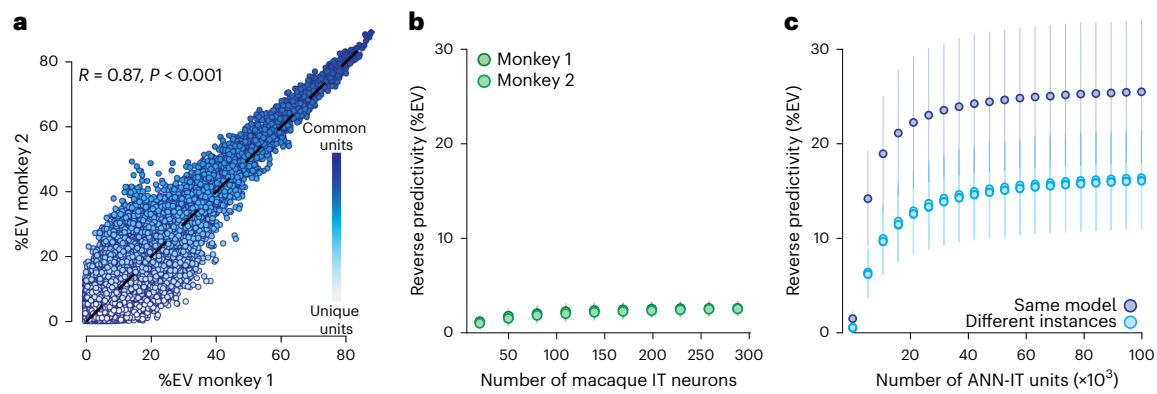


Fig. 5 | Unique units are consistent across monkeys and independent of neuron sampling. **a**, %EV of model units when predicted from each of two monkeys independently. Each dot represents a unit of ResNet-50. Unique units (light) are characterized by low %EV while common units (dark blue) show high %EV. Statistics were computed using Spearman correlation across 100,352 units of ResNet-50 ($P = 0.0$). **b,c**, Sampling analysis to predict unique units. Error

bars represent median absolute deviation across the 500 most unique units. **b**, Monkey-to-model reverse predictivity, as a function of sampled monkey neurons for either monkey. **c**, Reverse predictivity given units sampled from the same layer of ResNet-50 (dark blue) or the same layer of different instances of ResNet-50 (light blue).

Discussion

Our findings demonstrate a substantial asymmetry between current ANN models and the macaque IT cortex. While existing evaluations have focused on forward predictivity^{1,5}—how well model features can approximate neural responses. We demonstrate that this one-way comparison fails to identify truly brain-like models. The introduction of reverse predictivity, which quantifies how well IT neural activity can predict ANN unit activations, reveals that many model features remain untethered from any neural counterpart. This asymmetry was evident across model architectures and stands in contrast to the symmetry observed in monkey-to-monkey mappings, indicating that the discrepancy reflects genuine representational mismatch rather than noise or sampling limitations.

This asymmetry challenges the assumption that better forward predictivity implies that a model's internal representations are more biologically plausible. As we demonstrate (Figs. 1 and 2), two models can achieve comparable forward scores while relying on strikingly different strategies. Some have many units tightly aligned to neural activity, others have large pools of dimensions that serve no evident biological purpose. Of note, such mismatches are primarily informative when forward predictivity is already very high across multiple ANN classes. In this regime, the goal is no longer coarse model selection but finer-grained model falsification, or, more precisely, model discrimination, where these discrepancies help rule out otherwise similarly performing models. Reverse predictivity exposes these hidden differences, flagging the presence of units that, although useful for the model's task, are unaccounted for in the primate brain. The existence of such units suggests that current ANNs achieve high task performance with computational solutions that the brain may not implement, and which are not revealed by conventional benchmarks. These unique units are not simply the product of noise or poor sampling. We observed strong consistency in unit predictability across monkeys (Fig. 5a) and found that additional neurons did not improve the reverse predictivity of the least predictable units (Fig. 5b). Rather, these units are readily predicted by other ANN units, especially within the same architecture, indicating that they carry structured, yet model-specific, information (Fig. 5c). The brain's failure to predict them thus reflects not an inability to approximate model representations, but a divergence in the nature of those representations themselves. Reverse predictivity also offers a functional lens: units that are better aligned with neural activity tend to yield behavioural predictions that are more consistent across species and models. The units that are predictable

from IT neurons, common units, not only generalize better across architectures but also track primate behaviour more closely (Fig. 6). This suggests that the shared subspace between ANNs and the IT cortex captures the representational core necessary for object recognition³⁶. In contrast, unique units, although they may contribute to model performance (Extended Data Figs. 6 and 7), do not appear to support biologically plausible or behaviourally relevant computations. Identifying and prioritizing these shared components could thus refine both our understanding of biological vision and the development of more brain-like models.

Forward and reverse predictivity together reveal a tension: increases in one metric do not guarantee increases in the other. In fact, we observe that models with higher forward scores often fare worse in reverse predictivity, pointing to a growing divergence between ANN representations and the neural code as models scale in complexity and accuracy, something that the field has to contemplate further³⁷. This tension, however, does not imply an inherent trade-off. In some cases, both forward and reverse predictivity can be improved concurrently. For example, models jointly trained on object recognition and memorability exhibit increases in both metrics³⁸ (Extended Data Fig. 9). These results indicate that training objectives or architectural constraints that preserve forward predictivity while boosting reverse predictivity could be a promising direction for future work.

Factors such as feature dimensionality, training objective and architectural design all influence reverse predictivity (Fig. 4). Larger representational spaces tend to dilute neural alignment, while adversarial training improves it. Interestingly, SSL, despite its recent success, tends to yield lower reverse predictivity than supervised training. These insights suggest that reverse predictivity can serve not only as a benchmark but also as a constraint for future model development. Penalizing dimensions not predictable from neural activity may help enforce tighter alignment between artificial and biological systems. In addition, rather than optimizing only for task performance or forward neural predictivity, future models could be explicitly optimized to maximize ensemble common unit match, training ANNs such that their internal units become predictable from other ANN units. Such alignment-based training may help reduce idiosyncratic units and yield more brain-compatible models without sacrificing performance.

Reverse predictivity also complements previous approaches such as representational similarity analysis^{39,40}. While those methods capture global representational geometry or shared subspaces, reverse

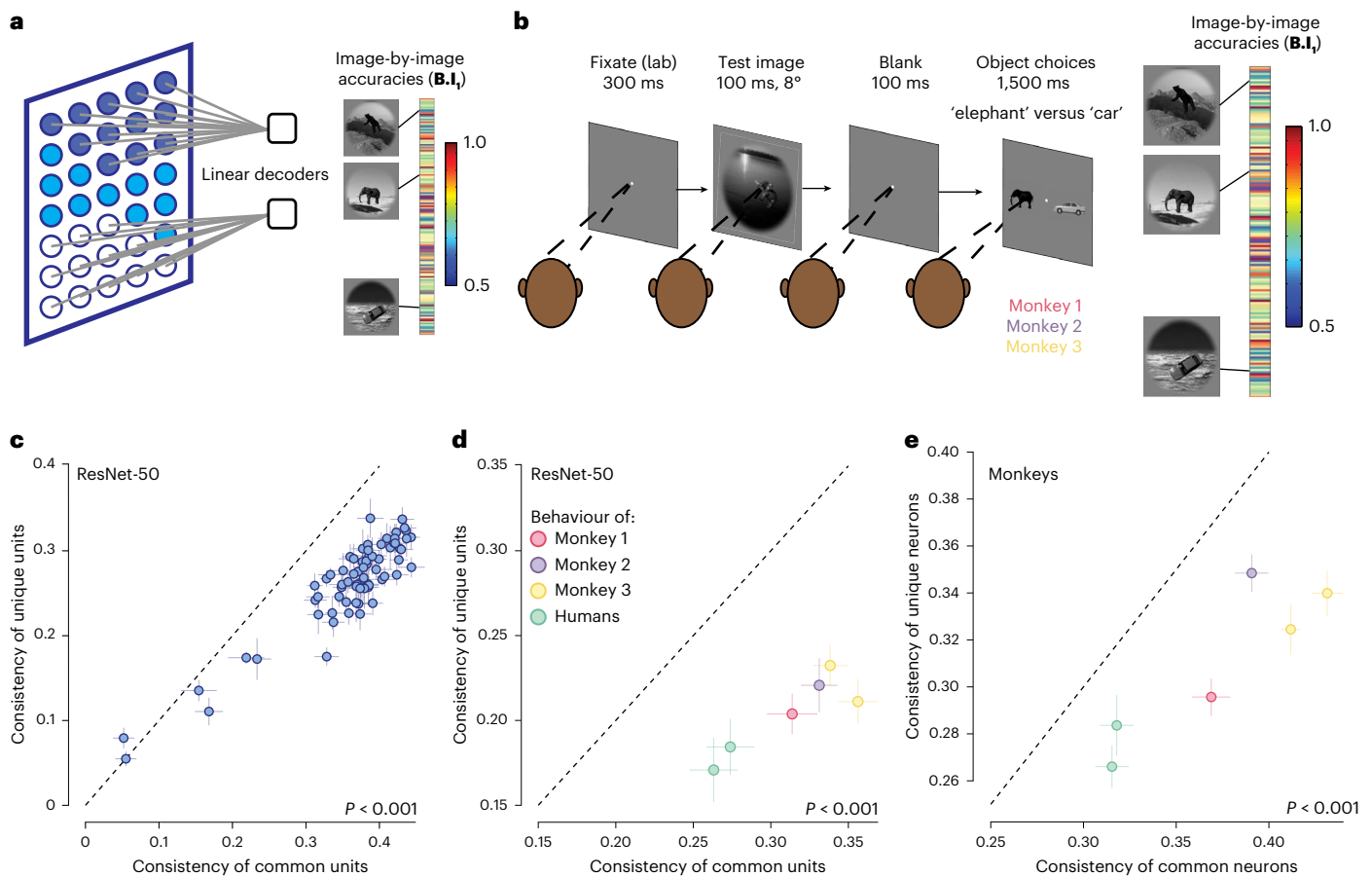


Fig. 6 | Common units drive consistent behaviour prediction across models and brains. **a**, Schematic of model-based behavioural prediction. Linear decoders are trained on common and unique units to predict object categories for each image, yielding an image-by-image behavioural signature. **b**, Schematic of the behavioural paradigm used with monkeys: after a brief fixation, test image and delay, animals choose between two object categories (for example, elephant versus car). Behavioural signatures are generated similarly for three monkeys (each shown in different colours corresponding to the datapoints used in **d** and **e**) and humans. **B.I.** is the image-by-image accuracy during the object discrimination task (Methods). **c**, Model-model consistency: consistency of common units versus unique units of ResNet-50 with all other models' behaviour. **d**, Model-to-brain consistency: consistency of common units versus unique

units of ResNet-50 with primate behaviour (three monkeys and humans). **e**, Monkey-to-monkey consistency: consistency of common neurons versus unique neurons of two monkeys with primate behaviour (same two monkeys as well as a third held-out monkey and humans). Error bars represent median absolute deviation. In **c–e**, the error bars represent median absolute deviation across 20 cross-validated bootstraps. Statistics were obtained using two-sided statistical paired tests, no adjustments for multiple comparisons (72 monkey-model pairs, Wilcoxon test, $P = 1.88 \times 10^{-13}$ (**c**); 6 comparisons (4 primate behaviours $\times$ ANN units selected from 2 primates; self-selection comparisons excluded) t -test, $P = 3.37 \times 10^{-5}$ (**d**); 6 comparisons (4 primate behaviours $\times$ neurons from 2 monkeys; within-primate excluded), t -test, $P = 7.27 \times 10^{-5}$ (**e**)). Images in **a** and **b** adapted from ref. 45, American Physiological Society.

predictivity directly quantifies whether the full span of model features is embedded within the brain's representational structure under a linear mapping assumption. The asymmetries we uncover suggest a many-to-one mapping: the brain captures only a subspace of model features, and the inverse mapping cannot predict the full model representation. This implies a non-invertible relationship, one in which ANN representations are richer or at least distinct from those in the IT cortex.

While our analyses focused on the IT cortex, it remains possible that some unique units correspond to computations in other brain regions⁴¹. However, the symmetry observed in monkey-to-monkey mappings, and the consistent behavioural alignment of common units, argues that the primary dimensions used for core object recognition are well captured by our IT recordings. Increasing the number of neurons or expanding the stimulus space might uncover additional alignments, but our sampling analyses suggest diminishing returns. Likewise, the assumption of linear mappings could be revisited, but if model features require nonlinear combinations of neural responses to be predicted, that itself implies a mismatch in representational format.

Similarly, although our study was grounded in vision, the conceptual framework is broadly applicable. In neuroscience domains where

ANN-brain comparisons are now commonplace, such as audition¹³, language¹⁵ and motor control⁴², reverse predictivity provides a principled tool to test whether model representations are truly embedded in the corresponding neural codes. Just as forward metrics have helped map task-driven ANNs onto sensory and motor areas, reverse predictivity can help ensure that such mappings are not only functionally relevant but structurally faithful. This is especially relevant as ANN-based models are increasingly used to interpret neural data and propose mechanistic hypotheses across cortical systems⁴³.

Ultimately, the goal of neuroscience-aligned modelling is not merely to match neural outputs but to mirror the brain's internal structure and computations³. Reverse predictivity advances this goal by assessing whether a model's internal code is embedded in neural activity at the level accessible to current linear analyses. Our findings demonstrate that no current ANN fully meets this criterion. Each carries representational baggage that the brain probably does not use. A model with high forward and reverse predictivity would reflect both behavioural performance and biological plausibility. We anticipate that reverse predictivity will become a critical target in the development and selection of next-generation models, guiding architectures towards

representations that not only solve tasks but also resonate with the brain's intrinsic structure. To facilitate systematic model comparison, we highlight that forward and reverse predictivity together define a natural two-dimensional space for assessing representational alignment. While one could construct a composite score to summarize performance in this space, the specific mathematical form of such a score is not essential to our conclusions. Instead, we emphasize the broader conceptual point: models that are both strong predictors of neural responses (forward predictivity) and themselves predictable from neural activity (reverse predictivity) satisfy a stricter and more biologically informative diagnostic criterion than either metric alone. Considering both dimensions jointly helps identify models whose representational structure is not only sufficient for predicting neurons but also embedded within the span of neural population activity. Future work may explore principled ways of combining or normalizing these metrics, potentially relative to brain-to-brain ceilings or dimensionality constraints to develop unified benchmarks for biological fidelity as ANN–brain comparisons continue to expand across sensory and cognitive domains.

Methods

Visual stimuli

We used 1,320 images previously used in a study⁸. The images contained ten object types (bear, elephant, face, apple, car, dog, chair, plane, bird, zebra). Images were either synthetic (but naturalistic) or natural photographs from the Microsoft Common Objects in Context (MS COCO) imageset⁴⁴. The synthetic images were rendered from a three-dimensional object model (originally purchased from TurboSquid) into a two-dimensional projection while varying an object's position (horizontal and vertical position), rotation (in the horizontal, vertical or depth plane) and size. The rendered object view was then added to a randomly chosen natural image background (indoor and outdoor scenes obtained from Dosch Design, www.doschdesign.com). All resulting images were greyscaled and had a resolution of 256×256 pixels. The MS COCO images were coloured but also contained the objects from the ten categories mentioned above. We used a high-resolution display with a refresh rate of 60 Hz for neural recordings and standardized luminance across images to control for low-level visual confounds.

To assess the generalizability of our results across visual domains, we introduced an additional set of 300 images. These images depict the same ten objects as before, but rendered in different styles—including outlines, drawings, schematized forms and artistic variations (see examples in Fig. 3b).

Subjects

Non-human primates. The non-human subjects in our experiments were three adult male rhesus monkeys (*Macaca mulatta*). Animals were outbred and therefore do not belong to defined strains or substrains. All animals were of Indian origin and were obtained from an Association for Assessment and Accreditation of Laboratory Animal Care-accredited institutional breeding colony. At the time of data collection, animals were aged 8–10 years. All surgical and animal procedures were performed in accordance with National Institutes of Health guidelines, the Massachusetts Institute of Technology Committee on Animal Care, and the Canadian Council on Animal Care on the use of laboratory animals and were also approved by the York University Animal Care Committee.

Large-scale neural recordings in the IT cortex

Data analysed in this article have been previously used for various scientific publications^{8,45,46}.

Surgical implants and microelectrode arrays. We surgically implanted three 10×10 microelectrode arrays (Utah Arrays, Blackrock Microsystems) per hemisphere in the IT cortex of each monkey

under aseptic conditions. Each array contained 96 electrodes, excluding corner electrodes, with a length of 1.5 mm and a spacing of 400 μm between electrodes. We determined the placement of the arrays intraoperatively using the visible sulcus patterns for guidance. For monkeys receiving implants in both hemispheres, we initially implanted arrays in one hemisphere and recorded data for approximately 1 year before explanting and reimplanting new arrays in the opposite hemisphere.

Electrophysiological recordings. During each experimental session, multiunit neural activity was recorded continuously at a sampling rate of 20 kHz using an Intan RHD Recording Controller (Intan Technologies). The raw voltage signals were bandpass filtered offline using a second-order elliptical filter (300 Hz to 6 kHz, 0.1 dB passband ripple, 50 dB stopband attenuation), before being thresholded to obtain the multiunit spike events. A multiunit spike event was defined as the threshold crossing when voltage (falling edge) deviated by more than three times the standard deviation of the raw voltage values.

The implanted arrays sampled a range of regions along the posterior-to-anterior axis of the IT cortex. For all analyses, we treated each recording site as a random sample from the broader IT population without considering the precise spatial locations of the electrodes. We analysed neural responses averaged between 70 ms and 170 ms after image onset.

Eye-tracking and calibration. During recording sessions, we monitored eye movements using video eye-tracking (SR Research EyeLink 1000). Using operant conditioning and water reward, the monkeys were trained to fixate a central white dot (0.2°) within a square fixation window that ranged from $\pm 2^\circ$. At the start of each behavioural session, monkeys performed an eye-tracking calibration task by making a saccade to a range of spatial targets and maintaining fixation for 500 ms. Calibration was repeated if drift was noticed over the course of the session.

Real-time eye-tracking was used to ensure that eye jitter did not exceed $\pm 2^\circ$, otherwise the trial was aborted, and data discarded. Stimulus display and reward control were managed using the MWorks Software (<https://mworks.github.io>).

Behavioural testing

Behavioural task. Each trial started with the presentation of a sample image (from a set of 1,320 images) for 100 ms at approximately 8° of visual angle. This was followed by a blank grey screen for 100 ms, after which the choice screen was shown, containing a standard image of the target object (the correct choice) and a standard image of the distractor object.

Non-human primates. In the behavioural experiment, monkeys performed an object recognition task. Images were presented on a 24-inch LCD monitor ($1,920 \times 1,080$ at 60 Hz) positioned 42.5 cm in front of the animal. Monkeys were head-fixed. Monkeys fixated on a white dot (0.2°) for 300 ms to initiate a trial. The monkey was allowed to view freely the choice objects for up to 1,500 ms and indicated its final choice by holding fixation over the selected object for 400 ms. Trials were aborted if gaze was not held within $\pm 2^\circ$ of the central fixation dot at any point until the choice screen was shown. Before testing in the laboratory, monkeys were trained in their home-cages⁴⁵ to perform the delayed match to sample tasks on the same object categories (but with a different set of images, see ref. 46 for details).

Humans tested via Amazon MTurk. Human behaviour (88 participants) was measured using the online Amazon MTurk platform, which enables efficient collection of large-scale psychophysical data from crowd-sourced human intelligence tasks (previously published⁸). Participants provided informed consent by clicking the 'Submit'

button on the first screen. The reliability of the online MTurk platform has previously been validated by comparing results obtained from online and in-lab psychophysical experiments^{35,47}. The participants indicated their choice by touching the screen or clicking the mouse over the target object. Each participant saw an image only once. Data were collected such that there were 80 unique participant responses per image with varied distractor objects.

Behavioural metrics. We used a one-versus-all image-level behavioural performance metric, **B.I_i**, similar to previous studies^{8,47,48}, to quantify the behavioural performance of monkeys and humans. This metric estimates the overall object discriminability of each image containing a specific target object from all other objects (pooling across all nine possible distractor choices).

As mentioned above, for each trial of the task, a specific sample image was shown and a binary choice task screen was presented. So the data obtained from each trial are a correct (1) or incorrect (0) choice from the participants. Each trial can be labelled with 2 unique identifiers—the unique sample image (one out of 1,320) and a unique task (1 out of 9 possible tasks given the image, for example bear versus dog).

Hence, given an image of object ‘i’, and all nine distractor objects ($j \neq i$), we computed the average performance per image (each element of the **B.I_i** vector) as the average of the percent correct across all the binary tasks done with that image as the sample image (where object ‘i’ was the target and all objects $j \neq i$ were the distractors respectively).

While the **B.I_i** vector provides an estimate of the monkeys’ image-by-image accuracy, the monkeys’ overall performance accuracy can be determined by taking an average of the **B.I_i** vector (across all images).

Analyses of ANN models

Model selection. We selected a broad and diverse set of 39 ANN models spanning a wide range of architectures, training objectives and inductive biases (see full list in Extended Data Fig. 10). These models include classic convolutional networks such as AlexNet²³, VGG16³² and multiple ResNet²⁴ variants (for example, ResNet-18, ResNet-50, ResNet-101, ResNet-152), as well as deeper architectures such as ConvNeXt²⁷, DenseNets³¹, Inception-v1/v3^{25,49} and NASNet/PNASNet^{50,51}. We also included lightweight models such as MobileNet^{52,53}, ShuffleNet⁵⁴ and SqueezeNet⁵⁵, biologically inspired models such as CORnet-S and CORnet-RT³⁶, and modern transformer-based architectures including ViT²⁶ and Swin transformers⁵⁷. To capture the effects of training, we incorporated models trained under standard supervised learning, SSL (for example, SimCLR⁵⁸, SwAV⁵⁹) and robust models developed by adversarial training⁶⁰. All models were pre-trained on the ImageNet³⁰ object classification dataset and task (except for minor differences in training specifics as per the original publications), and we used the publicly available pre-trained weights.

Layer selection. For each ANN, we extracted activations from a single high-level layer corresponding to the macaque IT cortex, as defined by the ANN-IT mapping on Brain-Score⁵. The layer selection in Brain-Score is done by asking which layer of the model best predicts existing macaque IT data, using data from a previous publication³⁵ and methods previously published⁴ (similar to our many-to-one regression method). Of note, we did not use the same neural data used in layer selection for our analyses in the article. Neural data⁸ used in this article were collected in a separate set of monkeys using a different image set. The number of units in the selected layer varies per model (for example, 4,096 for AlexNet, 100,352 for ResNet-50; see Extended Data Fig. 10 for a full list of models and their IT layers).

Effective dimensionality. To estimate the representational complexity of each model’s IT-layer features, we computed effective dimensionality using the participation ratio of the PCA eigenvalue spectrum. For each

model, given the activation matrix of units in the IT-mapped layer, we performed PCA to obtain the eigenvalues of the covariance matrix. The effective dimensionality was then defined as the participation ratio:

$$ED = \frac{(\sum_{i=1}^M \lambda_i)^2}{\sum_{i=1}^M \lambda_i^2}$$

where λ_i denotes the variance explained by the i th principal component.

Matched model comparisons. To isolate the effect of the training objective, we selected model pairs with identical architectures but differing in training strategy. For self-supervised^{61,62} versus supervised comparisons (Fig. 4e), we matched six models such as ResNet-50 trained with SimCLR⁵⁸ versus supervised ImageNet³⁰. For robustness comparisons (Fig. 4f), we used architectures trained with adversarial robustness objectives⁶³ and their standard counterparts. Reverse predictivity scores were averaged across units for each model and compared pairwise across models.

Analyses

Linear predictivity framework. All predictivity analyses used ridge regression with identical cross-validation (see ‘Cross-validation’ section) and reliability correction procedures. Depending on analysis direction, either neural responses or model features served as predictors.

Given predictor **X** and targets **y**, we estimated weights $w \in M \times 1$ and bias b to minimize the squared error (where M is the number of features in the predictor):

$$y_i \approx w^T \mathbf{X}_i + b$$

for image i (T refers to the transpose operation). Predictive performance was quantified as explained variance (%EV), defined as the squared Pearson correlation between predicted and observed responses, normalized by the geometric mean of their split-half reliabilities (see ‘Split-half reliability’), and scaled to percentage.

Forward predictivity analysis. To quantify forward predictivity, we evaluated how well each model’s features could linearly predict IT neural responses. We used only the most reliable neural sites (split-half reliability > 0.7). Then, for each model and each recorded neuron j , we fit a linear regression (see ‘Linear predictivity framework’) to predict the neuron’s responses across images from the model’s activations.

We computed %EV for each neuron, of each monkey and for each model. A model’s overall forward predictivity score was defined as the mean %EV across the full population of neurons.

Reverse predictivity analysis. Reverse predictivity measures how well the recorded IT neuron population can predict the activations of individual units within each model’s IT-layer representation. For each model unit, we treated its responses across images as the target variable and used neural responses as predictors.

We computed %EV for each model unit predicted from each monkey IT cortex. A model’s overall reverse predictivity score was defined as the mean %EV across all units. To characterize units by their reverse predictivity, we labelled those in the top 20th percentile as common units and those in the bottom 20th percentile as unique units.

Weight analysis. The linear weight vectors w_j obtained from the regression models indicate the contribution of each model unit to the prediction of individual IT neurons. To enable comparisons across models, we normalized (division by the maximum value) the weight

magnitudes within each model. We then assessed the distribution of weights for each unit across all neurons to identify units that were consistently influential versus those that were effectively ignored.

Sampling analysis of reverse predictivity. To examine how the number of sampled features affects the ability to predict unique units, we performed two complementary analyses. On the neural side (Fig. 5b), we randomly subsampled increasing numbers of IT neurons from each monkey (20 to 300, in steps of 20) and evaluated reverse predictivity for the 500 least predictable units (that is, lowest %EV) in the ResNet-50 IT layer. Each sample size was repeated 10 times, and the mean %EV across the 500 units was reported. On the model side (Fig. 5c), we sampled increasing numbers of units (1,000 to 100,000 in steps of 5,000) from the same ResNet-50 layer—excluding the 500 target units to be predicted—and used them to predict these same 500 units via linear regression. We tested two conditions: units sampled from the same model instance (dark blue) and units sampled across five independently initialized ResNet-50 instances (light blue).

Monkey-to-monkey predictivity. To estimate an upper bound on model–brain predictivity, we evaluated how well one monkey’s IT population could predict the other’s neural responses using the same linear regression framework used for model-based analyses. For each pair of monkeys, we treated one monkey’s IT responses as predictors and the other’s as targets. This analysis was conducted symmetrically in both directions (Monkey 1 → Monkey 2 and Monkey 2 → Monkey 1), and the resulting scores were averaged.

Model-to-model predictivity. To quantify the similarity between models’ internal representations, we measured how well one model’s features could predict another using the same linear regression framework. For each model pair, the source model’s feature vectors were treated as predictors and the target model’s features as responses, with the linear regression trained across images. Because model features are deterministic and noise-free, no reliability correction was applied. We conducted this analysis under two conditions. In the first, we assessed ResNet-50-to-ResNet-50 predictivity across multiple instances of the same architecture trained from different random initializations, estimating how consistently a fixed architecture converges to similar feature representations. In the second condition, we evaluated cross-predictivity across the full set of models used in the study (39 total), such that each model served as both predictor and target for every other model. This yielded $39 \times 38 = 1,482$ distinct prediction mappings, each evaluated symmetrically in both directions and averaged.

Feature-ablation test. To further evaluate the functional contributions of ANN units to forward and reverse predictivity measures, we performed a feature-ablation analysis across all 39 models included in the study. For each model, we first ranked units in the model’s IT layer by their reverse predictivity scores (%EV), computed on the additional dataset of 300 images (see ‘Visual stimuli’ and ‘Reverse predictivity analysis’ sections). Units within the top 20th percentile of this distribution were designated as common units, and units within the bottom 20th percentile were designated as unique units, consistent with the definitions used throughout the paper. To ensure independence from the subsequent ablation evaluation, this unit classification was performed using a cross-validation set that did not overlap with the images or neural responses used in the ablation tests. After identifying common and unique units, we ablated each group separately in features extracted from the corresponding held-out image set. Ablation was implemented by setting the activations of the designated units to zero before any subsequent computations. For each ablation condition (no-common, no-unique), we recomputed both forward predictivity and reverse predictivity using the same linear regression framework applied throughout the study (see ‘Linear predictivity framework’).

Extension of ablation test to behavioural readouts. We next evaluated the effects of common- and unique-unit ablations on behavioural-level model performance. From these ablated networks (defined above), we extracted features from the behavioural (penultimate) layer on a held-out behavioural stimulus set. For each model, we obtained two feature sets (IT-common–ablated and IT-unique–ablated) and evaluated both (1) image-level classification accuracy and (2) behavioural consistency with primate behaviour. Accuracy was computed on the image set used for the model-behaviour evaluations in the main text. Behavioural consistency was quantified as the correlation between model-level and primate-level behavioural signatures, computed for the human pool and each monkey.

Dimensionality-matched predictivity analyses. To evaluate whether limited neural sampling could account for the forward–reverse predictivity asymmetry, we performed dimensionality-matched analyses in which each model’s IT-layer representation was reduced to match the number of recorded IT sites ($N = 288$) using cross-validated PCA, in which principal components were estimated from ImageNet-validation features for that model and then applied to the experimental image set. All reduced representations were computed separately for each model and used in the predictivity analyses. Using these dimension-matched feature sets, we recomputed both forward and reverse predictivity as described above.

Consistency of common and unique units with behavioural responses. To assess the alignment between common and unique units and behavioural responses, units were first identified based on their reverse predictivity scores. Common units were defined as those in the top 20th percentile, and unique units as those in the bottom 20th percentile, based on predictions from both monkeys (or from the other monkey in monkey-to-monkey comparisons). To ensure comparability across models, each unit set was reduced to 1,000 dimensions using random projections.

To control for accuracy as a potential confound, common units were subsampled to match the accuracy of the unique units, which typically show lower accuracy. Specifically, subsets of common units (150–350 units) were evaluated, and the subset whose mean accuracy most closely matched that of the unique units was selected; if the unique-unit accuracy fell between two values, the lower-performing subset was chosen. This procedure was repeated with multiple random subsamples.

Behavioural responses from three monkeys and human participants were summarized as image-level response vectors (see ‘Behavioural testing’). For each ANN model, comparable image-level vectors were obtained by training cross-validated linear decoders on the model’s IT-layer activations to classify image object categories. Decoding was performed separately using (subsampled) common units and unique units, yielding two image-level response vectors per model or monkey.

Consistency between decoded responses and behavioural data from all other models, monkeys and human participants was quantified using Pearson correlation. Correlations involving primate behavioural data were corrected for measurement noise using split-half reliability (see ‘Split-half reliability of behavioural data’), and correlations involving neural data were corrected for neural recording noise using split-half reliability estimates specifically computed for common and unique neuron subsets (see ‘Neural split-half reliability for decoding’).

One-to-one mapping for forward and reverse predictivity. To evaluate one-to-one model–neuron correspondence (Extended Data Fig. 5), we used a cross-validated correlation-based mapping strategy (see ‘Cross-validation’). For each neuron, we identified the single best-predictive model unit using training data and measured its ability to predict test responses. Specifically, for each unit in the model’s IT-aligned layer, we computed its Pearson correlation with the training responses of the target neuron. The unit with the highest

correlation was selected as the best-predictive unit. We then measured the correlation between this unit's test responses and the test responses of the neuron. EV was computed as the squared test-set correlation, normalized by the split-half reliability of the neuron (see 'Split-half reliability'). This produced a %EV score, reported as the neuron's one-to-one forward predictivity.

For reverse predictivity, the procedure was inverted: for each model unit, we identified the best-predictive neuron using training data, then assessed how well that neuron could predict the unit's test responses. All other steps remained identical, including normalization, correlation-based selection and reliability correction (using the neuron's internal reliability). This one-to-one analysis provides a fine-grained, unit-level measure of alignment between model features and neural activity.

Statistical analyses

All statistical analyses were performed in Python using `scipy.stats`. Relationships between continuous variables (for example, reverse predictivity versus model complexity or effective dimensionality) were assessed using Spearman correlation coefficients unless otherwise noted.

Statistical testing. All statistical comparisons were guided by a consistent decision procedure based on the distributional properties and pairing of the data. First, we assessed whether each distribution was approximately normal using the Shapiro–Wilk test⁶⁴. If both groups satisfied the normality assumption, we applied parametric tests. For paired data (for example, within-participant or within-model comparisons), we used a paired *t*-test (two-tailed; degrees of freedom = $n - 1$). For unpaired comparisons (for example, between models), we used an independent samples *t*-test (degrees of freedom = $n_1 + n_2 - 2$).

If at least one distribution violated the normality assumption, we used non-parametric alternatives. Specifically, we applied the Wilcoxon signed-rank test for paired comparisons and the Wilcoxon rank-sum test (equivalent to the Mann–Whitney *U*-test) for unpaired data. All tests were two-tailed unless stated otherwise. We report exact *P* values and test statistics throughout.

Multiple comparisons and statistical interpretation. Our analyses consist of several conceptually independent comparisons (for example, model-to-brain versus monkey-to-monkey symmetry, architecture-specific trends, robustness comparisons, supervised versus self-supervised comparisons). Each comparison tests a distinct hypothesis and is therefore evaluated within its own family rather than pooled across the entire paper. Following common practice in computational and systems neuroscience, we do not apply corrections across unrelated analyses but instead report exact *P* values and test statistics for full transparency. Effects that do not reach significance are described as non-significant trends and are not used to support inferential claims.

Cross-validation. A tenfold cross-validation scheme was used, dividing the dataset into training and testing subsets. The dataset comprised 1,320 images distributed equally across 10 object categories. This design ensured a balanced representation of each category during training and testing.

Split-half reliability. Split-half reliability was used to estimate the internal consistency of neural and behavioural measurements. In all cases, data were partitioned into two independent halves by random resampling without replacement along the trial or repetition dimension.

For each split, responses were averaged across trials to yield mean responses. Depending on the analysis, these mean responses were then converted into image-level vectors (for example, **B.I₁** vectors) or neuron/unit-level predictions. Reliability was quantified as the Pearson

correlation between the two splits. To correct for the use of half the available data in each split, raw split-half correlations (*r*) were adjusted using the Spearman–Brown⁶⁵ formula:

$$\text{Corrected correlation} = \frac{2 \times r}{1 + r}$$

Split-half estimation was repeated multiple times with independent random splits, and the mean corrected correlation across repetitions was used as the final reliability estimate.

Split-half reliability of behavioural data. For behavioural data, split-half reliability was computed on the image-level behavioural performance metric (**B.I₁**). For each image, the minimum number of available trials across images was identified to ensure balanced sampling. These trials were randomly divided into two equal halves.

For each split-half, mean accuracy was computed per image, yielding two **B.I₁** vectors.

Neural split-half reliability. For neural data, split-half reliability was estimated by dividing neural responses along the trial (repetition) dimension into two independent halves. For each split, neural responses were averaged across trials to obtain mean firing rates per neuron and image.

Reliability is computed independently for every neuron, so no assumption is made that noise ceilings are identical across datasets or neurons. This neuron-specific correction ensures fair comparison across models even under heterogeneous noise conditions.

In model-to-brain and monkey-to-monkey analyses, %EV was normalized by the geometric mean of the split-half reliabilities of the predicted and target signals. When ANN models were used as predictors or targets, they were treated as deterministic, and a reliability of 1 was assumed.

Neural split-half reliability for decoding. To estimate the reliability of image-level decoding from neural data, we applied the split-half procedure described above to the decoding pipeline itself. Neural response data were randomly partitioned. For each split, neural responses were used to train and evaluate a decoding model, yielding two independent image-level decoded response vectors.

Decoded image-level vectors from the two splits were then correlated across images.

This decoding-specific reliability estimate was used when correcting correlations involving decoded neural representations, ensuring that reliability reflected variability arising from both neural response noise and decoder estimation.

Noise ceiling estimation. Noise ceilings were computed as the square root of the product of the relevant split-half reliabilities of the two variables being compared. For example, in comparisons between monkey and human **B.I₁** vectors, the noise ceiling was defined as:

$$\text{NC} = \sqrt{r_{\text{monkey}} \times r_{\text{human}}}$$

Data availability

To facilitate reproducibility, we uploaded the behavioural datasets, example model features, macaque IT neural responses and precomputed results necessary to reproduce all figures in the paper to Open Science Foundation (OSF): https://osf.io/y3qmk/?view_only=6dfe548c7ba24238932d247e65523053. The link is also included in the repository README and will remain permanently accessible.

Code availability

All data analysis was done using Python 3.11.13 and libraries `numpy` (1.26.4), `scipy` (1.15.3), `scikit-learn` (1.7.1), `matplotlib` (3.10.8) and `h5py` (3.14.0). To run the human behavioural experiments, we used

www.jpspsych.org/ version 6. All analysis code is available at https://github.com/vital-kolab/reverse_pred (<https://doi.org/10.5281/zenodo.17704686> (ref. 66)) under an MIT license. Detailed instructions for using the provided resources and running the analyses are documented in the repository's demo notebooks and README file. We also implemented a Python library (<https://pypi.org/project/reverse-pred/>) providing general and scalable tools to compute linear mappings between any population.

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Author contributions

S.M.: conceptualization, data curation, data analyses, writing—original draft, writing—review and editing, and visualization. K.K.: supervision, funding acquisition, conceptualization, data collection, data curation, writing—original draft, writing—review and editing, and visualization.

Competing interests

The authors declare no competing interests.

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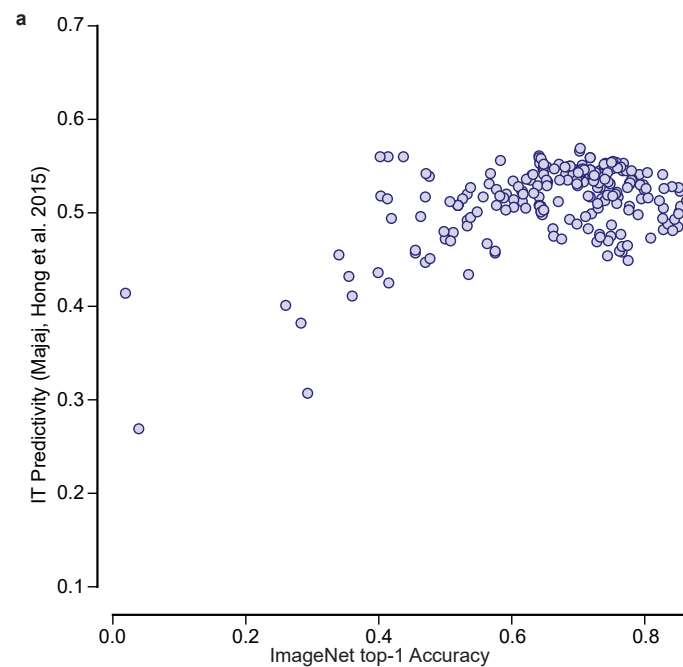
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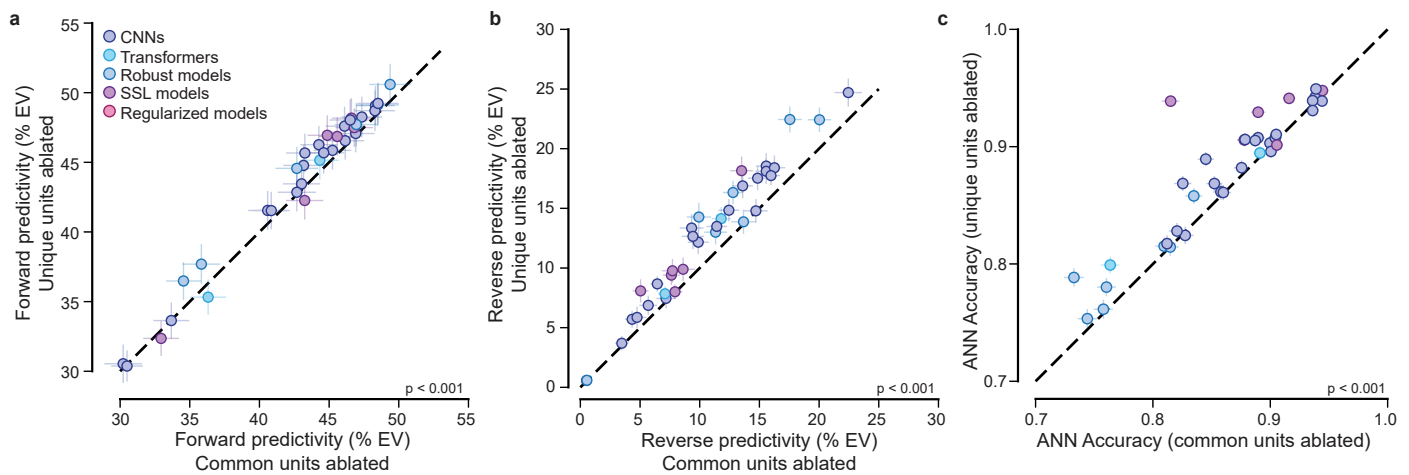
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**Extended Data Fig. 1 | ImageNet performance vs. forward predictivity.**

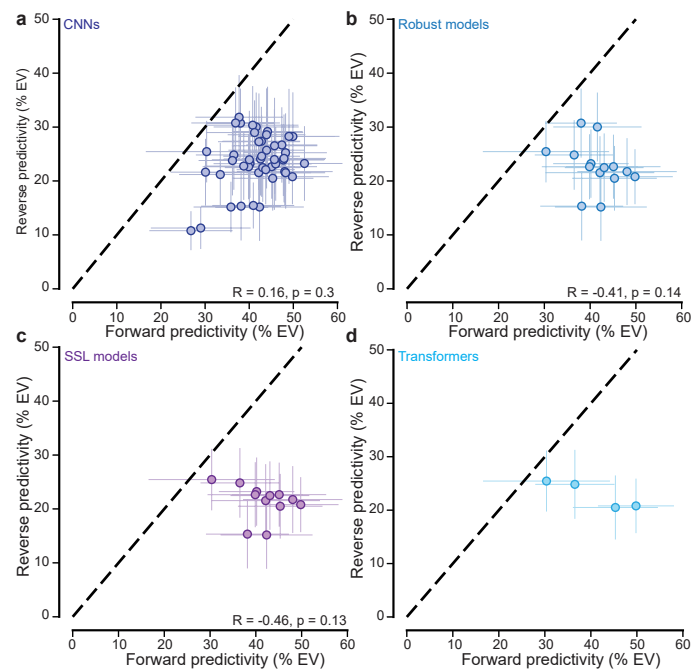
(a) Scatter plot showing the relation between ImageNet top-1 classification accuracy and forward predictivity for 254 ANN models in the MajajHong2015 IT benchmark³⁵. Each dot represents a model. Spearman correlation coefficient

(*Spearman* $r(252) = 0.15, p < 0.024$) indicates a moderate relationship between task performance and alignment with IT responses. However, as models get more accurate on ImageNet, their forward predictivity plateaus around 0.5 (that is approx. 50 % EV). Data for the plot is adapted from <http://brain-score.org>.



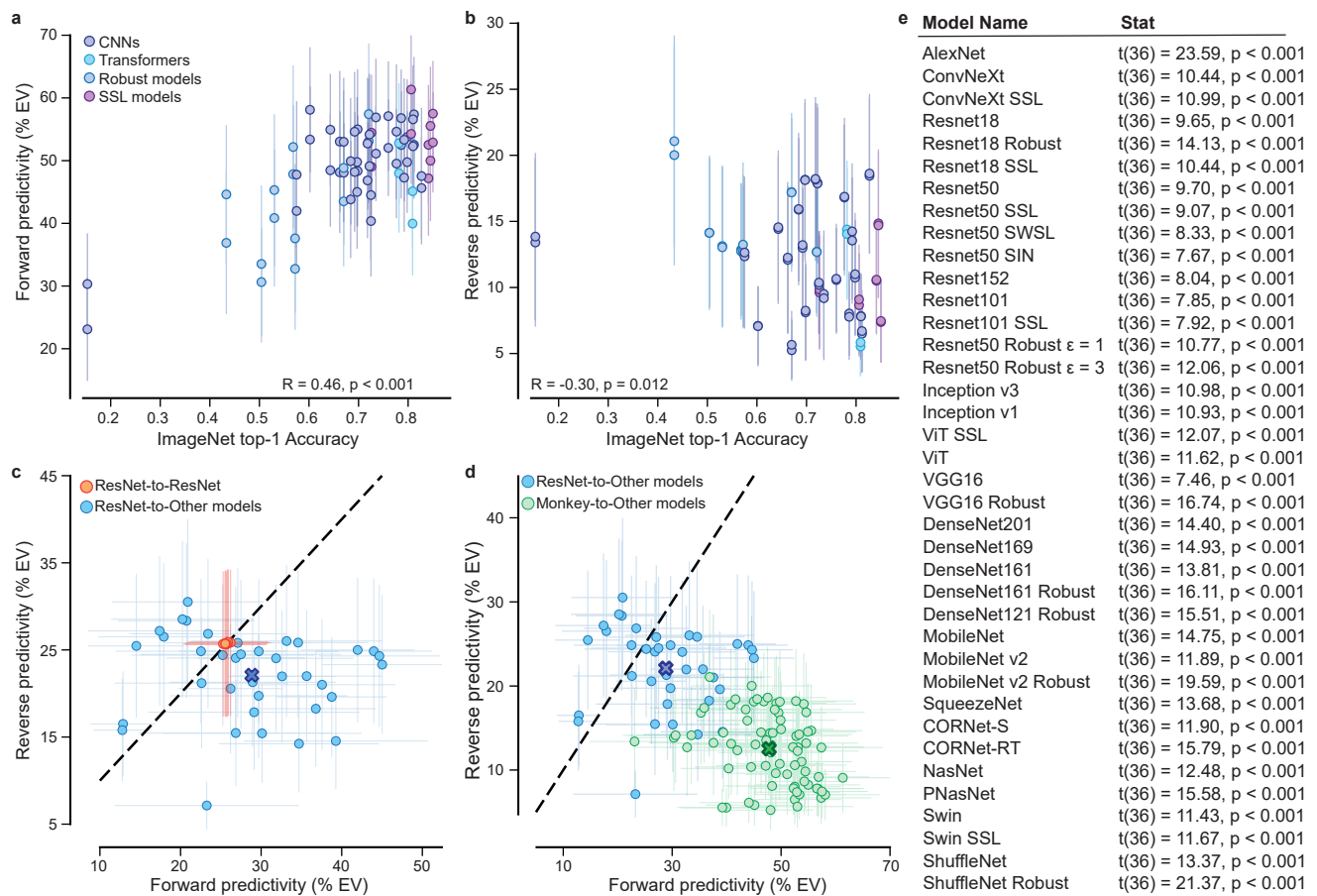
Extended Data Fig. 2 | Impact of common and unique units on bidirectional predictivity and model performance. (a) Forward predictivity per model when common units are ablated (x-axis) vs. unique units are ablated (y-axis). Even though unit types are defined by their respective reverse predictivity scores, removing common units decreases forward predictivity compared to removing unique units (36 models, two-sided paired wilcoxon test: $z = 63, p = 3.96 \times 10^{-66}$). Error bars represent Median Absolute Deviation across monkey neurons ($N_1 = 169$ and $N_2 = 107$). (b) Reverse predictivity per model when common units are ablated (x-axis) vs. unique units are ablated (y-axis). As expected and similar to (a), selectively removing common units decreases the reverse predictivity

score of each model, compared to removing unique units (36 models, two-sided paired wilcoxon test: $z = 0.0, p = 2.91 \times 10^{-11}$). Error bars represent Median Absolute Deviation across model units for each model (see number in Extended Data Fig. 10). (c) ANN Accuracy per model when common units are ablated (x-axis) vs. unique units are ablated (y-axis). Common and unique units are ablated from the IT layer while ANN performance is evaluated at the behavioural (penultimate) layer. Again, common unique have more impact on ANN accuracy compared to unique units (36 models, two-sided paired wilcoxon test: $z = 61, p = 3.19 \times 10^{-66}$). Error bars represent Median Absolute Deviation across 20 cross-validated bootstraps.



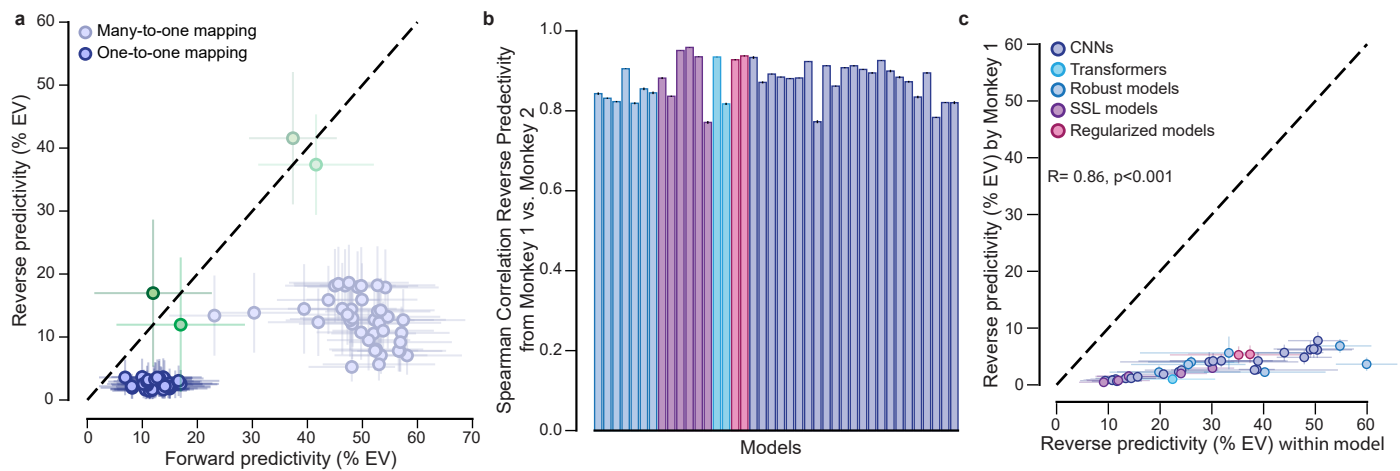
Extended Data Fig. 3 | Forward vs. Reverse Predictivity per model types after reducing the dimensionality of model features. The dimensionality of each ANN-IT layer is reduced using PCA (trained on a held-out feature set on the ImageNet dataset) to match the total number of IT neurons ($N = 288$). Forward and Reverse predictivity are re-computed on these new feature sets and plotted separately for CNNs (a), Robust model (b), Self-Supervised models (c) and transformers (d). In all cases, Forward predictivity scores remain significantly higher than Reverse predictivity scores ($p < 0.001$), and both metrics are not

correlated. (a–d) Error bars represent the Median Absolute Deviation across model units on the y-axis (see numbers in Extended Data Fig. 10) and monkey neurons on the x-axis ($N_1 = 169$ and $N_2 = 107$). Statistics were obtained using Spearman correlation, and two-sided paired tests with no adjustments for multiple comparisons (a: 46 models, wilcoxon test, $z = 0.0$, $p = 2.84 \times 10^{-14}$, b: 14 models, t-test, $t(13) = 11.7$, $p = 2.83 \times 10^{-08}$, c: 12 models, t-test, $t(11) = 11.1$, $p = 2.58 \times 10^{-07}$, d: 4 models).



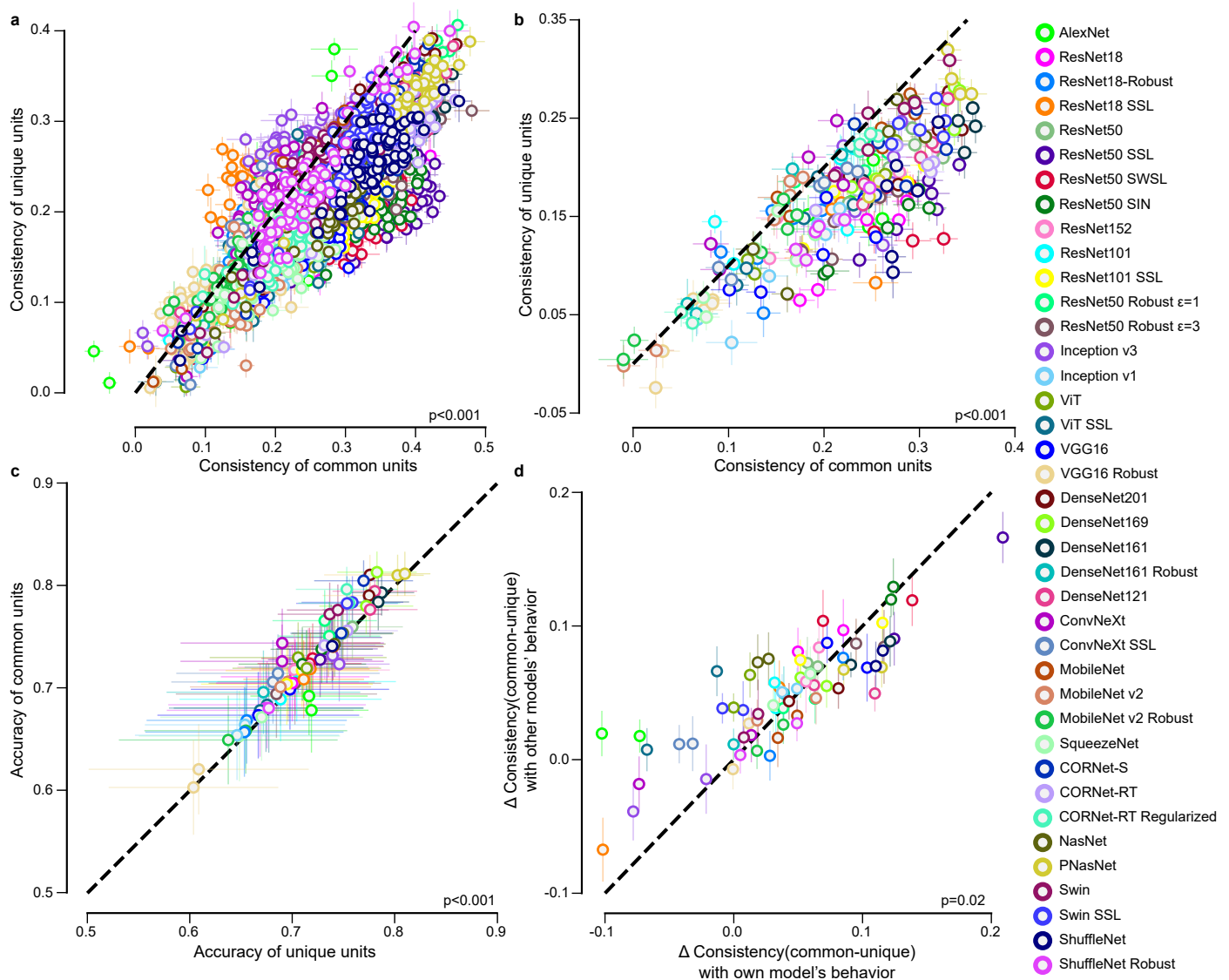
Extended Data Fig. 4 | Model-to-Model comparisons. **(a)** ImageNet accuracy vs. Forward Predictivity from our model selection (39 models). Similarly to Extended Data Fig. 1, we find a positive correlation between metrics (*Spearman* $r(78) = 0.46, p = 5.66 \times 10^{-05}$), with a plateau when ImageNet accuracy reaches above 0.6 (*Spearman* $r(57) = 0.11, p = 0.432$). Error bars represent Median Absolute Deviation across monkey neurons ($N_1 = 169$ and $N_2 = 107$). **(b)** ImageNet accuracy vs. Reverse Predictivity of 39 models. The trend is now reversed with a negative correlation (*Spearman* $r(78) = -0.30, p = 0.012$) between both metrics. Error bars represent Median Absolute Deviation across model units for each model (see number in Extended Data Fig. 10). **(a-b)** Statistics were obtained using Spearman correlation, across 78 model-monkey pairs. **(c)** Forward vs. Reverse predictivity between ResNet50 and other models (different initializations of ResNet50: orange dots, different architectures: blue dots). We observe high bidirectionality when systems are aligned by definition (same architecture but different initializations, paired test: $t(10) = -0.560, p = 0.587$ indicating no difference between forward and reverse predictivity scores). Additionally, when considering other architectures predicted by ResNet50 (forward predictivity, x-axis) or predicting ResNet50 (reverse predictivity, y-axis), we observe a deviance from the unity line (unpaired test between orange dots and blue dots:

$t(46) = 2.147, p = 0.037$), confirming that ResNet50 is less aligned to other models than to different initializations of ResNet50. **(d)** Forward vs. Reverse predictivity of ResNet50-to-Other models (blue dots, same as **(c)**) and Monkey-to-Other models (green dots). We observe that monkey-to-model predictions fall on average further to the unity line compared to ResNet50-to-models (paired test between green dots and blue dots: $t(38) = 9.7, p = 3.01 \times 10^{-09}$), indicating that ResNet50 remains more aligned to other models than macaque IT neurons. Error bars represent Median Absolute. **(e)** Quantitative results for all models comparing model-to-model predictivity vs. monkey-to-model. All models show a significant difference indicating that all models are more aligned to other models than to macaque IT neurons. **(d-e)** Statistics were obtained using two-sided t-tests, and no adjustments for multiple comparisons.



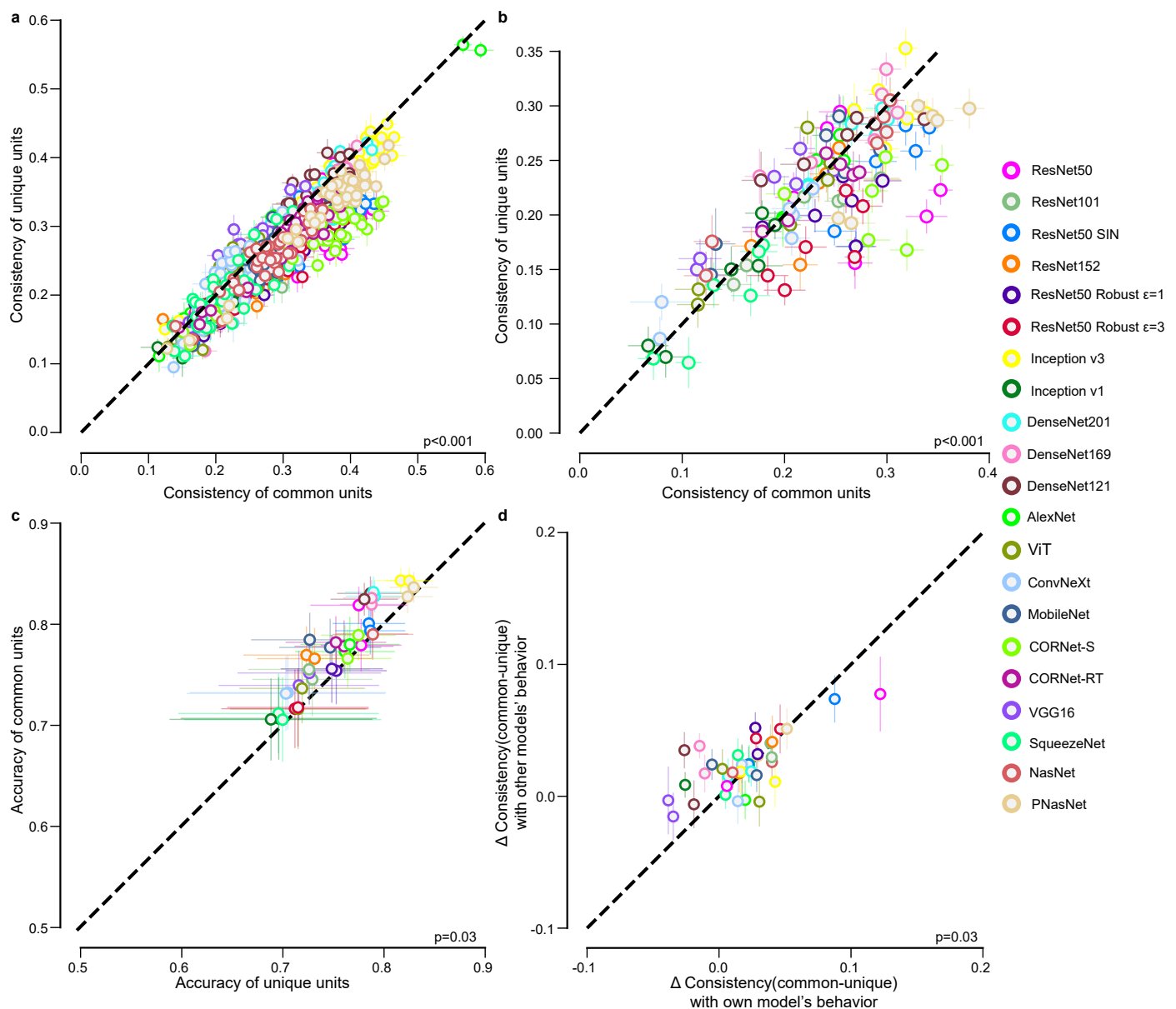
Extended Data Fig. 5 | One-to-one mapping and unit predictions across models. (a) Reverse predictivity (% explained variance) is plotted against forward predictivity for both one-to-one (dark dots) and many-to-one (light dots) mappings across models and monkeys. As expected, many-to-one mappings generally yield higher reverse EV than one-to-one mappings. However, the asymmetry remains between models and monkeys. Error bars represent Median Absolute Deviation across monkey neurons ($N_1 = 169$ and $N_2 = 107$) on the x-axis and across model units for each model (see number in Extended Data Fig. 10) on the y-axis. (b) Spearman correlations between unit-level reverse predictivity scores from two monkeys for each model. High correlations across all models indicate strong cross-subject consistency. Bars are color-coded

by model type: CNNs (dark blue), Transformers (light blue), robust models (cyan), self-supervised (purple), and regularized models (red). (c) Scatter plot comparing reverse predictivity (% EV) when unique model units are predicted by Monkey 1 neurons (y-axis) versus predicted by units of the same model and same layer (x-axis). Each dot represents a model, colored by model class: CNNs, Transformers, Robust, SSL, and Regularized models. Statistics were obtained using Spearman correlation across 78 model-monkey pairs ($r(76) = 0.86$, $p = 4.56 \times 10^{-07}$) indicating that unique units better predicted within models are also better predicted by monkeys. However, all models predict themselves better than monkey neurons do. Error bars represent Median Absolute Deviation across the 500 most unique model units for each model.



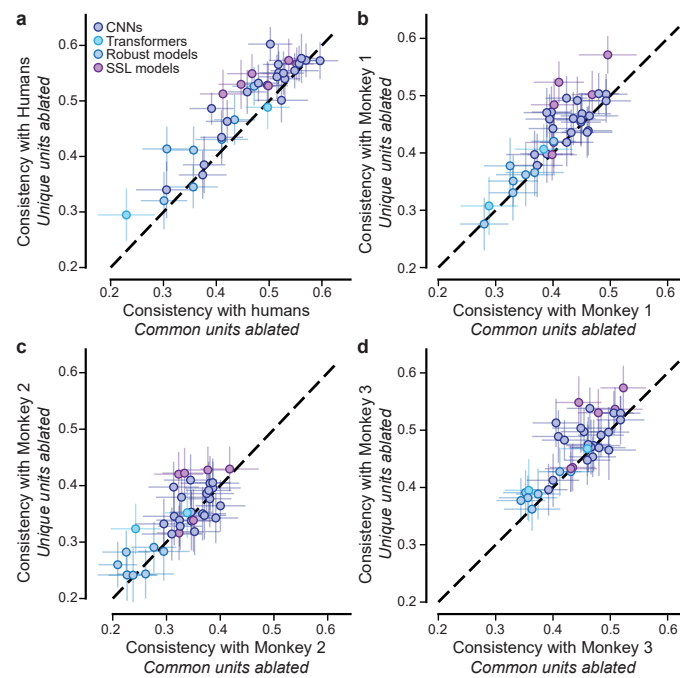
Extended Data Fig. 6 | Consistency of common vs. unique units in behavioural signatures. (a) For each model, the consistency of image-level behavioural predictions with all other models is plotted for common vs. unique units. Each cluster of colored points corresponds to a model instance. Across models, common units exhibit higher behavioural consistency than unique ones. (b) Same plot as (a), but showing the consistency with primates' (three monkeys and humans) image-level predictions. (c) Accuracy of common vs. unique units after subsampling common units to match, as closely as possible, the accuracy of the unique units. If the accuracy of the unique units falls between two subsampled values, we select the lower one for comparison. (d) Δ Consistency

between common and unique units, computed either with respect to the behaviour of the same model (within-model) or of all other models (between-model). The consistently smaller within-model deltas suggest that unique units retain some behavioural relevance specific to their source model. (a-d) Error bars represent Median Absolute Deviation across 20 cross-validated bootstraps. Statistics were obtained using two-sided paired tests, with no adjustments for multiple comparisons (a: 2808 model-model pairs, wilcoxon test, $z = 245628$, $p = 0.0$, b: 234 model-primates comparisons, wilcoxon test, $z = 919$, $p = 3.56 \times 10^{-35}$, c: 78 model-monkey pairs, t-test, $t(77) = -6.31$, $p = 1.64 \times 10^{-08}$, d: 78 model-monkey pairs, t-test, $t(77) = -2.08$, $p = 0.02$).



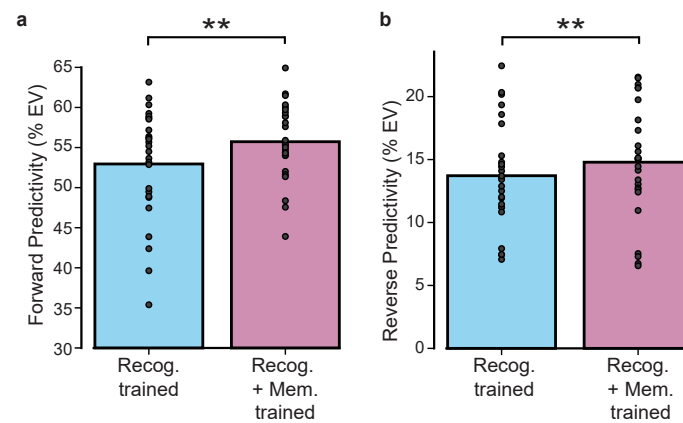
Extended Data Fig. 7 | Consistency of common vs. unique units in behavioural signatures from one-to-one mapping. (a) For each model, the consistency of image-level behavioural predictions with all other models is plotted for common vs. unique units, selected best on reverse predictivity from one-to-one mappings. Each cluster of colored points corresponds to a model instance. Across models, common units exhibit higher behavioural consistency than unique ones. (b) Same plot as (a) but showing the consistency with primates' (three monkeys and humans) image-level predictions. (c) Accuracy of common vs. unique units after subsampling common units to match, as closely as possible, the accuracy of the unique units. If the accuracy of the unique units falls between two subsampled values, we select the lower one for comparison. (d) Δ Consistency

between common and unique units, computed either with respect to the behaviour of the same model (within-model) or of all other models (between-model). The consistently smaller within-model deltas suggest that unique units retain some behavioural relevance specific to their source model. (a-d) Error bars represent Median Absolute Deviation across 20 cross-validated bootstraps. Statistics were obtained using two-sided paired tests, with no adjustments for multiple comparisons (a: 798 model-model pairs, wilcoxon test, $z = 38403$, $p = 5.02 \times 10^{-77}$; b: 126 model-primates comparisons, wilcoxon test, $z = 2638$, $p = 0.0009$; c: 42 model-monkey pairs, t-test, $t(41) = -8.29$, $p = 2.68 \times 10^{-10}$; d: 42 model-monkey pairs, t-test, $t(41) = -1.10$, $p = 0.03$).



Extended Data Fig. 8 | Impact of unique types on consistency with primate behaviour. Similarly to Extended Data Fig. 2, we selectively ablate common and unique units from the ANN-IT layer and compare the ANN's behaviour with measured primates' behaviour (N = 36 models, **a**: humans, **b-d**: monkeys),

Consistent with results from Extended Data Fig. 2, removing common units alters the model's consistency with primate behaviour compared to removing unique units. Error bars represent Median Absolute Deviation across 20 cross-validated bootstraps.



Extended Data Fig. 9 | Joint training on recognition and memorability improves both reverse and forward predictivity. We compared 26 ANNs only for recognition vs. the same architectures trained jointly for recognition and memorability. We observe that the jointly trained ANNs outperform the recognition ANNs on forward predictivity (a), but also reverse predictivity (b).

This implies that while both metrics seem negatively correlated (see Fig. 3), it is possible to design models that show improvement on both metrics. Each dot represents a model (N = 26 models). Statistics were obtained using one-sided paired t-tests, with no adjustments for multiple comparisons (a: $t(25) = -2.803$, $p = 0.005$, b: $t(25) = -3.146$, $p = 0.004$). **: $p < 0.01$.

Model Name	Architecture Type/Objective	ANN-IT Layer	# units
AlexNet	CNN	features[12]	4096
ConvNeXt	CNN	features[5][4]	75264
ConvNeXt SSL	CNN-SSL	stages[2].blocks[4]	75264
Resnet18	CNN	layer4[0].bn1	25088
Resnet18 Robust	CNN-Robust	layer4[0].bn1	25088
Resnet18 SSL	CNN-SSL	layer4[0].bn1	25088
Resnet50	CNN	layer4[0].relu	100352
Resnet50 SSL	CNN-SSL	layer4[0].relu	100352
Resnet50 SWSL	CNN-SSL	layer4[0].relu	100352
Resnet50 SIN	CNN-Stylized ImageNet ⁵⁴	layer4[0].relu	100352
Resnet152	CNN	layer3[34].bn3	200704
Resnet101	CNN	layer3[22].relu	200704
Resnet101 SSL	CNN-SSL	layer3[22].relu	200704
Resnet50 Robust $\epsilon = 1$	CNN-Robust $\epsilon = 1$	layer4[0]	100352
Resnet50 Robust $\epsilon = 3$	CNN-Robust $\epsilon = 3$	layer4[0]	100352
Inception v3	CNN	Mixed_7a	32000
Inception v1	CNN	inception4a	100352
ViT	Transformer	encoder.layers[6]	151296
ViT SSL	Transformer-SSL	encoder.layers[6]	151296
VGG16	CNN	features[23]	100352
VGG16 Robust	CNN-Robust	features[23]	100352
DenseNet201	CNN	features.transition3.pool	43904
DenseNet169	CNN	features.transition3.pool	31360
DenseNet161	CNN	features.transition3.pool	51744
DenseNet161 Robust	CNN-Robust	features.transition3.pool	51744
DenseNet121 Robust	CNN-Robust	features.transition3.pool	25088
MobileNet	CNN	features[12]	10976
MobileNet v2	CNN	mobilenet_v2.layer[12]	18816
MobileNet v2 Robust	CNN-Robust	mobilenet_v2.layer[12]	18816
SqueezeNet	CNN	features[9]	32448
CORNet-S	CNN	IT.t0	25088
CORNet-RT	RNN	IT.t4	25088
CORNet-RT reg	RNN-regularized ¹²	IT.t4	25088
NasNet	CNN	cell_12	197568
PNasNet	CNN	cell_8	211680
Swin	Transformer	layers[3].blocks[0].norm2	37632
Swin SSL	Transformer-SSL	layers[3].blocks[0].norm2	37632
ShuffleNet	CNN	stage4	22736
ShuffleNet Robust	CNN-Robust	stage4	22736

Extended Data Fig. 10 | List of models with their ANN-IT layers. Exhaustive list of all the models used in this study, along with their architecture types and training objectives (if nothing is specified, Supervised Learning on ImageNet is employed), as well as their ANN-IT layer and the number of units in each of these layers.